

TICVAI

FINANCE BACKEND STRUCTURE

Module Reference | Version 1.0

A connected finance architecture covering setup, ledger, revenue, reconciliation, refunds, cash management, reporting, controls, and AI.

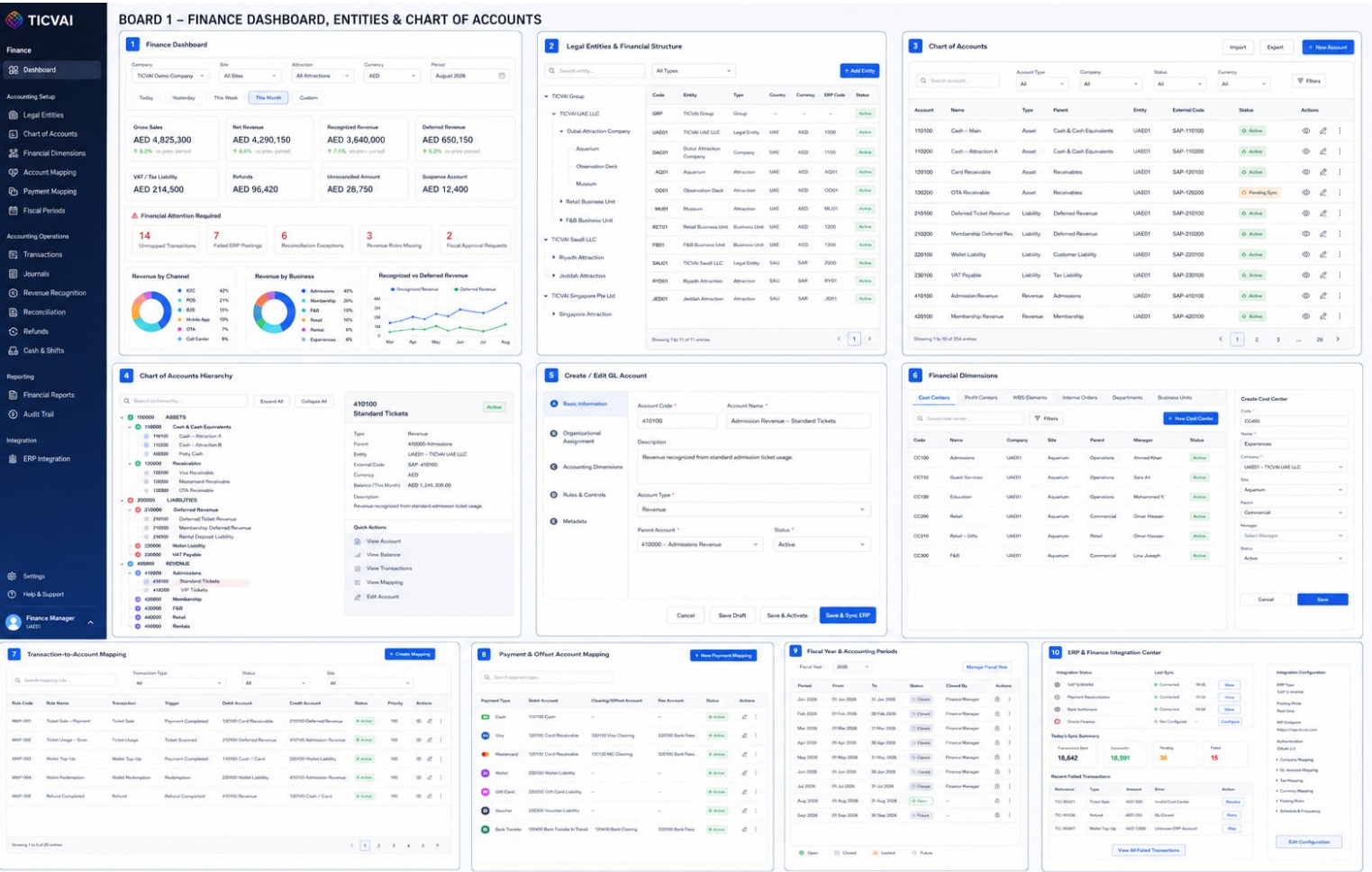
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Finance module boards and their starting pages

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Document structure: seven connected boards forming a single enterprise finance layer.

BOARD 1 — Finance Dashboard, Entities & Chart of Accounts



Purpose of the board: Establish the financial structure that every TICVAI transaction will use later for journals, revenue recognition, reconciliation, reporting and ERP posting.

The overall hierarchy should be:

Legal Entity → Business Unit → Attraction/Site → Department → Financial Dimension → GL Account → Transaction Mapping

Screen 1 — Finance Dashboard

This is the main landing page when a Finance Manager enters the Finance module.

Top section

Global filters:

Company: TICVAI Demo Company

Site: All Sites

Attraction: All Attractions

Currency: AED

Period: August 2026

Also:

Today | Yesterday | This Week | This Month | Custom

Main KPI cards

First row:

Gross Sales

AED 4,825,300

↑ 8.2% vs previous period

Net Revenue

AED 4,290,150

Recognized Revenue

AED 3,640,000

Deferred Revenue

AED 650,150

Second row:

VAT / Tax Liability

AED 214,500

Refunds

AED 96,420

Unreconciled Amount

AED 28,750

Suspense Account

AED 12,400

Operational finance alerts

Separate section:

Financial Attention Required

- 14 Unmapped Transactions
- 7 Failed ERP Postings
- 6 Reconciliation Exceptions
- 3 Revenue Rules Missing
- 2 Fiscal Approval Requests

Each number should be clickable.

Charts

Revenue by Channel

- B2C
- POS
- B2B
- Mobile App
- OTA
- Call Center

Revenue by Business

- Admissions

- Membership
- F&B
- Retail
- Rental
- Experiences

Recognized vs Deferred Revenue

Monthly trend.

Important recommendation

This dashboard should not just show financial reports. It should act as an **exception-management dashboard**.

Finance should immediately see:

“What needs my attention?”

instead of only:

“How much did we sell?”

Screen 2 — Legal Entities & Financial Structure

This screen defines the organizational structure used by Finance.

Left-side hierarchy

Example:

TICVAI Group

- ↳ TICVAI UAE LLC
 - ↳ Dubai Attraction Company
 - ↳ Aquarium
 - ↳ Observation Deck
 - ↳ Museum
- ↳ TICVAI Saudi LLC
 - ↳ Riyadh Attraction
 - ↳ Jeddah Attraction

Main table

Columns:

Code	Entity	Type	Country	Currency	ERP Code	Status
UAE01	TICVAI UAE LLC	Legal Entity	UAE	AED	1000	Active
AT01	Aquarium	Attraction	UAE	AED	AQ01	Active
RET01	Retail	Business Unit	UAE	AED	1200	Active

Add Financial Entity

Fields:

- Entity Name

- Entity Code
- Entity Type
- Parent Entity
- Country
- Base Currency
- Tax/VAT Registration No.
- ERP Company Code
- Fiscal Calendar
- Default Chart of Accounts
- Default Cost Center
- Default Profit Center
- Status

Entity types

Group

Legal Entity

Business Unit

Attraction

Venue

Site

Department

Sales Office

The matrix specifically requires financial separation and reporting across multiple companies, sites and business entities.

Screen 3 — Chart of Accounts

The main COA management screen.

Header

Chart of Accounts

Buttons:

+ New Account

Import

Export

Bulk Update

Sync ERP

Filters

Search account...

Account Type

Company

Site

Status

Currency

External ERP Code

Main table

Account	Name	Type	Parent	Entity	External Code	Status
110100	Cash – Main	Asset	Cash	UAE01	SAP-110100	Active
120100	Card Receivable	Asset	Receivables	UAE01	SAP-120100	Active
210100	Deferred Ticket Revenue	Liability	Deferred Revenue	UAE01	SAP-210100	Active
220100	Wallet Liability	Liability	Customer Liability	UAE01	SAP-220100	Active
410100	Admission Revenue	Revenue	Admissions	UAE01	SAP-410100	Active

Account statuses

Active

◦ Inactive

Pending ERP Sync

ERP Error

Row actions

View

Edit

Clone

Deactivate

View Transactions

View Balance

View Audit

Screen 4 — Visual Chart of Accounts Hierarchy

Instead of only viewing accounts as rows, Finance can see the entire structure visually.

Example

100000 — ASSETS

↳ 110000 — Cash & Cash Equivalents

↳ 110100 Cash – Attraction A

↳ 110200 Cash – Attraction B

↳ 110300 Petty Cash

↳ 120000 — Receivables

↳ 120100 Visa Receivable

- ↳ 120200 Mastercard Receivable

- ↳ 120300 OTA Receivable

200000 — LIABILITIES

- ↳ 210000 Deferred Revenue

- ↳ 210100 Deferred Ticket Revenue

- ↳ 210200 Membership Deferred Revenue

- ↳ 210300 Rental Deposit Liability

- ↳ 220000 Wallet Liability

- ↳ 230000 VAT Payable

400000 — REVENUE

- ↳ 410000 Admissions

- ↳ 410100 Standard Tickets

- ↳ 410200 VIP Tickets

- ↳ 420000 Membership

- ↳ 430000 F&B

- ↳ 440000 Retail

- ↳ 450000 Rentals

Functions

Expand / Collapse

Drag into hierarchy — **only if authorized**

Add Child Account

View Account

View Balance

View Mapping

The matrix allows parent-child structures, account grouping, multi-level subaccounts, roll-ups and up to 10-level hierarchy.

Screen 5 — Create / Edit GL Account

This should be a detailed drawer or full backend screen.

Section A — Basic Information

Account Code

410100

Account Name

Admission Revenue – Standard Tickets

Description

Revenue recognized from standard admission ticket usage.

Account Type

Revenue

Parent Account

410000 – Admissions Revenue

Status

Active

Section B — Organizational Assignment

Company

Business Unit

Site

Attraction

Department

Sales Office

Location

Allow multiple assignments if permitted.

Section C — Accounting Dimensions

GL Account

Cost Center

Profit Center

Internal Order

WBS Element

Business Unit Code

External Accounting Code

ERP Account Code

This follows the ERP/COA requirements in the matrix.

Section D — Rules

Allow Manual Posting?

Yes / No

Allow Balance Transfer?

Yes / No

Allow Revenue Recognition?

Yes / No

Allow Negative Balance?

Yes / No

Requires Approval?

Yes / No

Section E — Metadata

Tags

Notes

Effective From

Effective To

Created By

Last Updated

ERP Sync Status

Buttons

Save Draft

Save & Activate

Save & Sync ERP

Screen 6 — Financial Dimensions: Cost Centers, Profit Centers & WBS

I recommend one common **Financial Dimensions** page rather than separate disconnected screens.

Tabs:

Cost Centers | **Profit Centers** | **WBS** | **Internal Orders** | **Departments** | **Business Units**

Example — Cost Center tab

Code	Name	Company	Site	Parent	Manager	Status
CC100	Admissions	UAE01	Aquarium	Operations	Ahmed	Active
CC110	Guest Services	UAE01	Aquarium	Operations	Sara	Active
CC200	Retail	UAE01	Aquarium	Commercial	Omar	Active

Create Cost Center

- Code
- Name
- Description
- Company
- Site
- Attraction
- Parent
- Validity Date

- ERP Code
- Manager
- Status

Why this matters

Products should not manually contain accounting logic.

For example:

Aquarium Ticket

can inherit:

Site = Aquarium

Profit Center = Aquarium Admissions

Cost Center = Guest Operations

from configured financial rules.

Screen 7 — Transaction-to-Account Mapping

This is probably the **most important configuration screen on Board 1**.

The system needs configurable mappings between operational events and financial accounts. The matrix explicitly requires mappings by product, transaction, channel, location and other conditions.

Header

Accounting Mapping Rules

+ Create Mapping

Table

Rule	Transaction	Trigger	Debit	Credit	Status
MAP-001	Ticket Sale	Payment	Card Receivable	Deferred Revenue	Active
MAP-002	Ticket Usage	Scan	Deferred Revenue	Admission Revenue	Active
MAP-003	Wallet Top-Up	Payment	Cash/Card	Wallet Liability	Active
MAP-004	Wallet Usage	Redemption	Wallet Liability	Revenue	Active
MAP-005	Refund	Refund Complete	Revenue	Cash/Card	Active

Create Mapping Rule

Rule Name

Standard Ticket Sale

Transaction Type

Ticket Sale

Trigger

Payment Completed

Conditions

Product = Standard Ticket

Channel = All

Site = Aquarium

Customer Type = Any

Payment Method = Card

Accounting Result**Debit**

Card Receivable Account

Credit

Deferred Ticket Revenue

Tax

VAT Payable

Priority

100

Effective Dates

From / To

Important

Mappings need:

- Rule priority
- Effective dates
- Versioning
- Clone
- Audit
- Approval
- Conflict detection

Screen 8 — Payment & Offset Account Mapping

This screen focuses specifically on payment methods.

The matrix specifically requires offset/balancing accounts by payment type.

Payment methods

Cash

Visa

Mastercard

Amex

Apple Pay

Google Pay

Digital Wallet

Gift Card

Voucher

Bank Transfer

OTA

Reseller

Internal Credit

Table

Payment Type	Company	Debit Account	Clearing Account	Cost Center	Status
Cash	UAE01	Cash	—	Admissions	Active
Visa	UAE01	Card Receivable	Visa Clearing	Admissions	Active
Mastercard	UAE01	Card Receivable	MC Clearing	Admissions	Active
Wallet	UAE01	Wallet Liability	—	—	Active
Voucher	UAE01	Voucher Liability	—	—	Active

Detailed configuration

Payment Provider

Acquiring Bank

Merchant ID

Terminal / POS Group

GL Account

Clearing Account

Fee Account

Commission Account

Tax Account

ERP Code

Currency

Settlement Frequency

This will later connect directly with reconciliation.

Screen 9 — Fiscal Year & Accounting Period Management

Page header

Fiscal Calendar

Selected:

FY 2026

Periods

Period	From	To	Status	Closed By
Jan 2026	01 Jan	31 Jan	Closed	Finance Manager
Jul 2026	01 Jul	31 Jul	Closed	Finance Manager
Aug 2026	01 Aug	31 Aug	Open	—
Sep 2026	01 Sep	30 Sep	◦ Future	—

Actions

Open

Close

Lock

Unlock

Reopen

Closing workflow

When Finance clicks:

Close August 2026

system validates:

All journals balanced

Revenue jobs completed

⚠ 7 reconciliation exceptions

⚠ 2 transactions in suspense

ERP posting completed

Then:

Submit Period Closing for Approval

Finance Manager

→ CFO

Important rule

If a period is closed, ordinary users cannot post transactions into that period.

If reopened:

- reason mandatory
- approval mandatory
- full audit trail

The matrix explicitly requests open/close/lock/unlock/reopen control with audit tracking.

Screen 10 — ERP & Finance Integration Center

This is where Finance/IT can see the integration between TICVAI and external finance systems.

Cards**SAP S/4HANA**

Connected

Last Sync: 19:35

Payment Reconciliation

Connected

Bank Settlement

Connected

Oracle Finance

◦ Not Configured

Integration statisticsTransactions Sent Today: **18,642**Successful: **18,591**Pending: **36**Failed: **15****Failed transactions**

Reference	Type	Amount	Error	Action
TIC-90421	Ticket Sale	AED 500	Invalid Cost Center	Resolve
TIC-90436	Refund	AED 250	GL Closed	Retry
TIC-90497	Wallet	AED 1,000	Unknown ERP Account	Map

Actions

Retry

View Payload

Correct Mapping

Push Manually

Exclude

Download Error

Configuration area

ERP Type

API Endpoint

Authentication

Company Mapping

GL Mapping

Tax Mapping

Currency Mapping

Posting Frequency:

- Real-time
- Near Real-time
- Scheduled Batch
- End of Day

The matrix supports API delivery, real-time posting APIs, ERP synchronization and financial exports.

Board 1 Navigation

For the actual design, I recommend the Finance sidebar look like:

Finance

Dashboard

Accounting Setup

- Legal Entities
- Chart of Accounts
- Financial Dimensions
- Account Mapping
- Payment Mapping
- Fiscal Periods

Accounting Operations

- Transactions
- Journals
- Revenue Recognition
- Reconciliation
- Refunds
- Cash & Shifts

Reporting

- Financial Reports
- Audit

Integration

- ERP Integration

One addition I recommendation

There should be a **financial configuration inheritance model**.

For example, instead of configuring the same GL mapping 500 times:

Group Default

↓ overridden by

Legal Entity

↓ overridden by

Site / Attraction

↓ overridden by

Product Group

↓ overridden by

Individual Product

So TICVAI can have:

Default Admission Revenue = 410000

while Aquarium can override it to:

Aquarium Admission Revenue = 410100

and one special VIP product could override it again to:

VIP Admission Revenue = 410150

This will make the Finance configuration much more manageable for large attractions and enterprise clients.

BOARD 2 — Transactions, Journals & Double-Entry Ledger

BOARD 2 – TRANSACTIONS, JOURNALS & DOUBLE-ENTRY LEDGER

1 Financial Transaction Explorer

Transactions Today	Gross Value	Posted	Pending
18,642	AED 4,825,300	18,501	96

2 Transaction Financial Detail

Transaction: TIC-2026-0090421 (Posted)

Type: Ticket Sale, PNR: 874251, Date: 11 Aug 2026 14:21, Channel: B2C Website, Site: Aquarium, Customer: Guest 5421, Payment: Visa, Gateway Ref: PG-8214517

3 Journal Entry List

Posted Today	Pending Approval	Drift	Reversed
124	12	8	3

4 Journal Entry Detail

Journal Reference: JE-100421 (Posted)

Journal Date: 11 Aug 2026, Posting Date: 11 Aug 2026, Source: Ticket Sale

Transaction: TIC-90421, PNR: 874251, Company: UAE01, Site: Aquarium, Currency: AED, ERP Reference: SAP-8745102

5 Create Manual Journal Entry

Base Information: Company: UAE01 - TICVAI UAE LLC, Journal Date: 11 Aug 2026, Posting Date: 11 Aug 2026

Journal Type: Adjustment, Reference: AED-REV-025

Description: Regularly revenue from admission to experience

6 Journal Approval Center

Pending My Approval: 12, Total Value: AED 842,500, High Risk: 3, Overdue: 1

7 Reversal, Reclassification & Transfer

Reversal Journal: JE-100421 - Ticket Sale

8 Suspense & Accounting Exceptions

Supervise Amount: AED 125,400, Transactions: 47, Critical: 6, Days: 8

9 General Ledger Explorer

Account: 47000 - Admission Revenue

Period: August 2026

10 Financial Transaction Audit & Trail

PNR: 874251 (Completed)

Purpose: Give Finance complete visibility and control from the original operational transaction through accounting treatment and eventual ERP posting.

The basic flow is:

Operational Transaction → Financial Transaction → Accounting Rule → Journal → Debit/Credit → GL → Approval → Posting → ERP → Audit

Screen 1 — Financial Transaction Explorer

This should be the **master search screen** for every financial transaction generated anywhere in TICVAI.

Header

Financial Transactions

Top KPI cards:

Transactions Today

18,642

Gross Value

AED 4,825,300

Posted

18,501

Pending

96

Exceptions

31

Failed

14

Search

One global search field:

Search PNR, Order, Ticket, Invoice, Payment, Journal...

A Finance user should be able to paste almost any TICVAI reference into this field.

Filters

- Date
- Company
- Site
- Attraction
- Transaction Type
- Source
- Channel
- Payment Method
- Currency
- Journal Status
- ERP Status

Transaction types

Examples:

Ticket Sale

Ticket Redemption

Membership

F&B Sale

Retail Sale

Rental

Wallet Top-up

Wallet Redemption

Gift Card

Refund

Discount

Deposit

Commission

Chargeback

Main table

Reference	Type	Source	Customer	Amount	Payment	Journal	ERP
TIC-90421	Ticket Sale	B2C	Guest 5421	AED 525	Visa	Posted	Synced
TIC-90422	F&B Sale	POS	Guest 2871	AED 85	Cash	Posted	Synced
TIC-90423	Wallet Top-up	App	Guest 9012	AED 500	Apple Pay	Posted	Pending
TIC-90424	Refund	Call Center	Guest 2214	-AED 250	Visa	Pending	—

Row actions

View Transaction

View Journal

View Payment

View Customer

View Order

View ERP Status

This page should become the Finance team's **financial Google search** for TICVAI.

Screen 2 — Transaction Financial Detail

When Finance clicks any transaction from Screen 1, they arrive here.

This is one of the most important screens in the entire Finance module.

Example**Order TIC-2026-0090421****Transaction Type:** Ticket Sale**PNR:** PNR-874251**Date:** 11 Aug 2026 – 14:21**Channel:** B2C Website**Site:** Aquarium**Customer:** Guest 5421**Payment:** Visa**Gateway Reference:** PG-8214517**Transaction summary**

Ticket Value: AED 500

VAT: AED 25

Total Paid: AED 525**Products**

2 × Aquarium Adult Ticket

AED 250 each

Accounting treatment

The system should visually explain what happened.

STEP 1 — SALE

Payment received

↓

Debit

Card Receivable

AED 525

↓

Credit

Deferred Ticket Revenue

AED 500

Credit

VAT Payable

AED 25

Status: Posted

Then later:

STEP 2 — REDEMPTION

Ticket scanned at 16:32

↓

Debit

Deferred Ticket Revenue

AED 500

↓

Credit

Admission Revenue

AED 500

Status: Posted

This gives Finance complete traceability from **sale → payment → ticket → scan → revenue**.

Tabs

Overview

Accounting

Payment

Revenue Recognition

Reconciliation

ERP

Audit

Screen 3 — Journal Entry List

This is the central journal-management screen.

Header

Journal Entries

KPI cards:

Posted Today

Pending Approval

Draft

Reversed

Failed

ERP Pending

Filters

Journal Reference

Date

Company

Journal Type

Source

Account

Cost Center

Status

Created By

ERP Status

Journal types

Sales

Revenue Recognition

Refund

Wallet

Membership

Deposit

Adjustment

Manual

Reversal

Reclassification

Commission

Table

Journal	Date	Source	Reference	Debit	Credit	Status	ERP
JE-100421	11	Ticket Sale	TIC-90421	525	525	Posted	Synced

Journal	Date	Source	Reference	Debit	Credit	Status	ERP
	Aug						
JE-100422	11 Aug	Ticket Scan	TIC-90421	500	500	Posted	Synced
JE-100423	11 Aug	Refund	TIC-90424	250	250	Pending	—
JE-100424	11 Aug	Manual	MAN-102	5,000	5,000	Approval	—

Important rule

Every journal must show:

Total Debit = Total Credit

If not:

UNBALANCED — POSTING BLOCKED

Your matrix explicitly requires validation that total debits equal total credits.

Screen 4 — Journal Entry Detail

This is the accounting-level view of one journal.

Journal JE-100421

Status: Posted

Header information

Journal Date: 11 Aug 2026

Posting Date: 11 Aug 2026

Source: Ticket Sale

Transaction: TIC-90421

PNR: PNR-874251

Company: UAE01

Site: Aquarium

Currency: AED

Created By: System

ERP Reference: SAP-8745102

Journal lines

Line	Account	Description	Debit	Credit	Cost Center	Profit Center
1	120100	Visa Receivable	525	—	—	AQ01
2	210100	Deferred Ticket Revenue	—	500	—	AQ01
3	230100	VAT Payable	—	25	—	AQ01

Bottom**Total Debit: AED 525****Total Credit: AED 525****Balanced****Quick actions**

View Original Transaction

View Revenue Event

View ERP Posting

Download

Reverse Journal

Audit Trail

Critical design principle

Finance should always be able to move backwards:

Journal → Transaction → Order → Product → Payment → Customer

and forwards:

Journal → GL → ERP → Reconciliation

Nothing should become an isolated accounting record.

Screen 5 — Create Manual Journal Entry

Although most journals should be automatically generated, Finance needs controlled manual journals.

Your matrix specifically requires manual journal creation, account transfers, revenue reclassification, transaction dates, descriptions, unique references and debit/credit entries.

Header**Create Manual Journal**

Automatically generated:

Journal Reference: MJ-2026-00158**Basic information**

Company *

Journal Date *

Posting Date

Currency *

Journal Type

Reason *

Description *

Reference

Attachment

Journal lines

Account	Debit	Credit	Cost Center	Profit Center	Description
410100 Admission Revenue	10,000	—	—	AQ01	Reclassification
410200 VIP Revenue	—	6,000	—	AQ01	Reclassification
410300 Experience Revenue	—	4,000	—	AQ01	Reclassification

Bottom:

Debit AED 10,000

Credit AED 10,000

Difference AED 0

Balanced

Buttons

Save Draft

Validate

Submit for Approval

Users should **not** have a simple "Post" button unless their permission/approval policy explicitly allows it.

Screen 6 — Journal Approval Center

This should connect with TICVAI's overall approval engine rather than Finance building a completely separate approval system.

Header

Financial Approvals

KPI:

12 Pending My Approval

AED 842,500 Total Value

3 High Risk

Approval queue

Request	Type	Amount	Requested By	Reason	Risk	Age
MJ-158	Manual Journal	10,000	Finance Executive	Reclassification	Medium	25m
MJ-159	Adjustment	125,000	Finance Manager	Year-end correction	High	2h
REV-91	Reversal	42,500	Finance Executive	Wrong account	High	1h

Open request

Show:

Original transaction

Proposed accounting

Before

↓

Proposed Change

↓

After

This is important because an approver should understand **what financial impact they are approving**, not just see "Approve / Reject."

Actions

Approve

Reject

Return for Correction

Add Comment

Approval workflow

Example:

Finance Executive

↓

Finance Manager

↓

CFO

Rules could depend on value:

Below AED 10,000 → Finance Manager

AED 10,000–100,000 → Finance Manager + Controller

Above AED 100,000 → CFO

Screen 7 — Journal Reversal, Reclassification & Transfer

This screen handles corrections.

A core accounting principle should be:

Posted financial transactions are never deleted or silently edited.

Your matrix explicitly states that posted financial records should be immutable.

Select action

Reverse

Reclassify

Transfer

Example — Reclassification

Original:

Admission Revenue

AED 20,000

Finance discovers it should have been:

VIP Revenue

AED 20,000

The system creates:

Original journal

JE-100500

↓

Reversal

JE-100601

↓

Corrected Journal

JE-100602

Fields

Reason *

Effective Date *

Destination Account *

Amount *

Description *

Attachment

Options

Full amount

Partial amount

Audit relationship

The UI should visibly show:

JE-100500

↕ linked to

REV-100601

↕ linked to

JE-100602

So an auditor can immediately understand the correction.

Screen 8 — Suspense & Accounting Exceptions

This screen catches transactions TICVAI cannot correctly account for.

It is essential because a sophisticated financial engine should **never guess where money belongs**.

Header**Suspense & Exceptions**

KPIs:

Suspense Balance: AED 125,400

Transactions: 47

Critical: 6

Oldest: 8 Days

Reasons**Missing Account Mapping — 14****Invalid Cost Center — 9****ERP Account Rejected — 7****Unknown Payment Type — 6****Revenue Allocation Error — 5****Closed Accounting Period — 4****Other — 2****Table**

Transaction	Amount	Issue	Suspense Account	Age	Owner
TIC-90455	5,000	Missing GL	999999	2h	Finance
TIC-90462	12,500	Invalid Cost Center	999999	5h	Finance
TIC-90321	42,000	ERP Rejection	999999	2d	Finance

Open exception

Show:

Original transaction

Failed rule

Expected mapping

System error

Suggested resolution

Actions

Assign Account

Correct Mapping

Reclassify

Retry

Escalate

Once resolved:

Suspense

→ Correct Account

→ Corrective Journal

→ ERP Retry

The matrix specifically requires automatic routing of unmapped revenue into suspense accounts, administrator alerts, manual reclassification and suspense reconciliation.

Screen 9 — General Ledger Explorer

This is where Finance examines account activity and balances.

Header

General Ledger

Select:

Company: UAE01

Period: August 2026

Account:

410100 — Admission Revenue

Account summary

Opening Balance

AED 12,400,000

Debits

AED 340,000

Credits

AED 4,850,000

Closing Balance

AED 16,910,000

Activity table

Date	Journal	Transaction	Description	Debit	Credit	Balance
11 Aug	JE-100421	TIC-90421	Admission Revenue	—	500	16,900,500
11 Aug	JE-100431	TIC-90431	Admission Revenue	—	250	16,900,750
11 Aug	REV-10045	TIC-90382	Refund/Reversal	100	—	16,900,650

Filters

Date

Company

Site

Account

Account Type

Cost Center

Profit Center

Currency

Journal Type

Transaction Source

Drill-down

Clicking any AED 500 entry should open:

GL Entry → Journal → TICVAI Transaction

This is particularly useful for audits and month-end investigation.

Screen 10 — Financial Transaction Audit & Traceability

This should be the final page of Board 2 and one of TICVAI's strongest Finance screens.

Search

Enter:

PNR-874251

Then TICVAI generates the entire financial timeline.

Example timeline

11 Aug — 14:21:03

Order Created

TIC-90421

↓

14:21:06

Visa Payment Authorized

AED 525

↓

14:21:07

Payment Captured

↓

14:21:07

Journal JE-100421 Generated

↓

14:21:08

✓ Debit/Credit Validation Passed

↓

14:21:09

Sent to ERP

↓

14:21:11

✓ ERP Accepted

SAP Reference 8745102

↓

16:32:14

Ticket Scanned

↓

16:32:15

Revenue Recognition Triggered

↓

16:32:15

Journal JE-100422 Generated

↓

16:32:17

✓ ERP Posted

Audit details

For every step show:

- Timestamp
- User/System
- Action
- Previous Value
- New Value
- Device/source
- Approval
- Reference
- Related transaction

The matrix requires transaction traceability, journal traceability, change history, reconciliation history, approval history and complete financial auditing.

Board 2 Navigation

The navigation should now look like:

Finance

Dashboard

Accounting Setup

- Legal Entities
- Chart of Accounts
- Financial Dimensions
- Account Mapping
- Payment Mapping
- Fiscal Periods

Transactions & Ledger

- Financial Transactions
- Journal Entries
- Manual Journals
- Financial Approvals
- Suspense & Exceptions
- General Ledger

Revenue

- Revenue Recognition

Reconciliation

- Payment Reconciliation

Reporting

- Financial Reports
- Audit & Traceability

One important recommendation for Board 2

We would recommend creating a **single Financial Event model** behind all these screens.

Do **not** let them build separate accounting logic inside POS, Ticketing, F&B, Retail, Membership, Wallet, Rental, etc.

Instead:

Ticketing / POS / F&B / Retail / Wallet / Membership / Rental

↓

Financial Event

Source + Transaction + Entity + Amount + Currency + Tax + Product + Payment + Timestamp

↓

Accounting Rules Engine

↓

Double-Entry Ledger

↓

Revenue Recognition

↓

Reconciliation

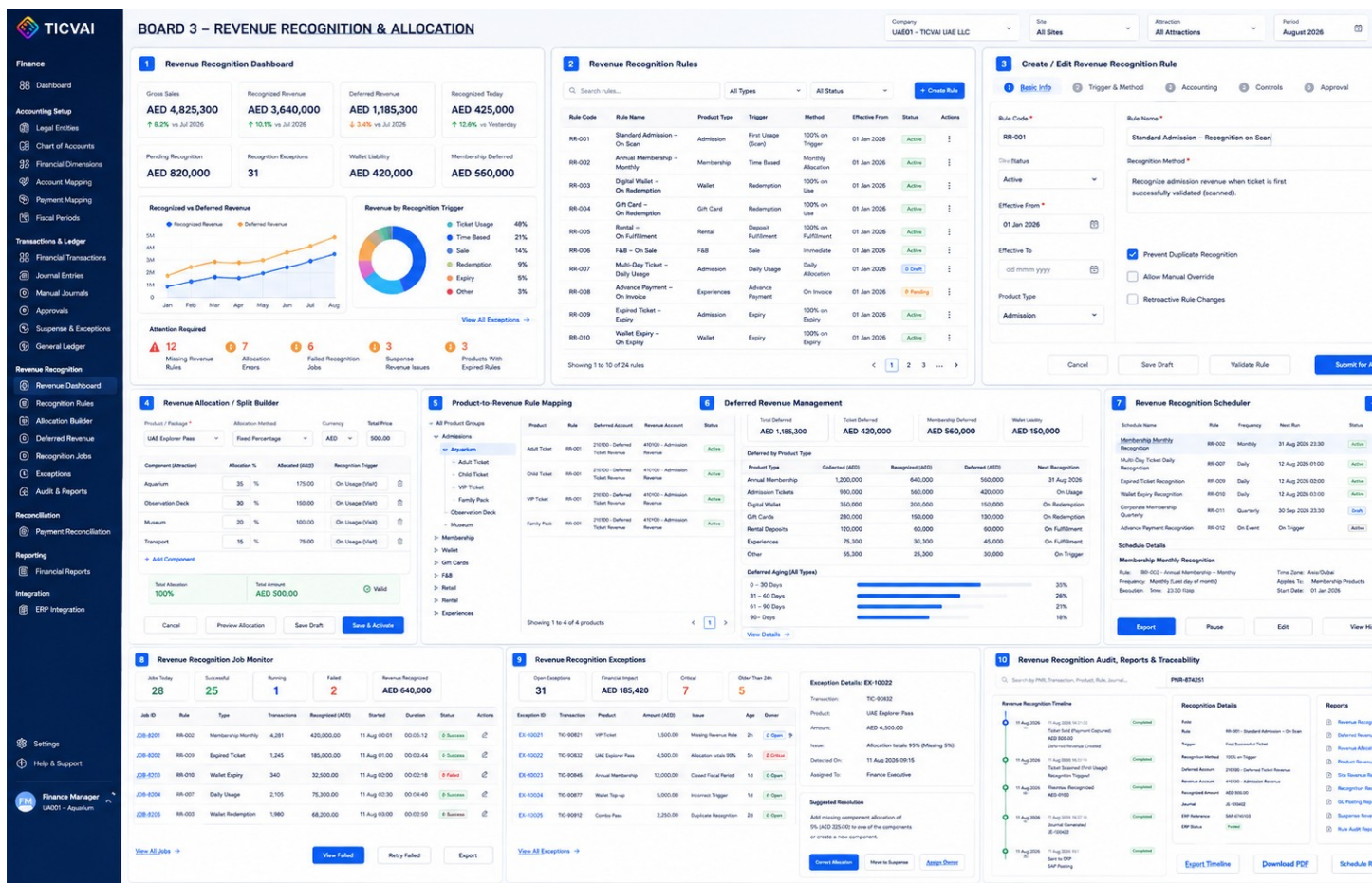
↓

ERP

This becomes especially important later. For example, when TICVAI adds a new module, the developers should not need to rebuild Finance. The new module simply produces standardized **Financial Events**, and the Finance Engine determines the accounting treatment.

That architecture, combined with the mapping configuration from Board 1, is what will make TICVAI's Finance layer scalable rather than just a collection of accounting screens.

BOARD 3 — Revenue Recognition & Allocation



Purpose

Board 2 answered:

“What financial transaction happened?”

Board 3 answers:

“When does TICVAI actually recognize the money as revenue, how much belongs to each product/site/entity, and where should it be posted?”

The basic engine should be:

Sale → Deferred Revenue → Recognition Trigger → Allocation → Earned Revenue → GL → ERP

Screen 1 — Revenue Recognition Dashboard

This is the Finance Manager's main overview of recognized and deferred revenue.

Global filters

At the top:

Company: UAE01

Site: All Sites

Attraction: All

Product: All

Revenue Rule: All

Currency: AED

Period: August 2026

KPI cards

Gross Sales

AED 4,825,300

Recognized Revenue

AED 3,640,000

Deferred Revenue

AED 1,185,300

Recognized Today

AED 425,000

Pending Recognition

AED 820,000

Recognition Exceptions

31

Wallet Liability

AED 420,000

Membership Deferred

AED 560,000

Main chart

Recognized vs Deferred Revenue

Monthly graph:

Jan → Aug

Two lines/bars:

Recognized Revenue

Deferred Revenue

Revenue by trigger

Donut/chart:

Ticket Usage — 48%

Time Based — 21%

Sale — 14%

Redemption — 9%

Expiry — 5%

Other — 3%

Revenue by business

Admissions

Membership

F&B

Retail

Rental

Wallet

Gift Card

Experiences

Attention Required

A separate section:

- 12 Missing Revenue Rules

- 7 Allocation Errors

- 6 Failed Recognition Jobs

- 3 Suspense Revenue Issues

- 3 Products With Expired Rules

Clicking any number opens the exception list.

Recommendation

Like Board 1, don't make this only a reporting dashboard.

It should answer:

“What revenue requires Finance attention today?”

Screen 2 — Revenue Recognition Rules

This is the master list of all revenue recognition rules.

Header

Revenue Recognition Rules

Buttons:

+ Create Rule

Clone Rule

Bulk Assignment

Export

Search & filters

Search Rule...

Trigger Type

Product Type

Company

Site

Status

Effective Date

Main table

Rule	Product Type	Recognition Trigger	Method	Effective	Status
RR-001	Admission	Ticket Scan	On Usage	01 Jan 26	Active
RR-002	Membership	Time	Monthly	01 Jan 26	Active
RR-003	Wallet	Redemption	On Usage	01 Jan 26	Active
RR-004	Gift Card	Redemption	On Usage	01 Jan 26	Active
RR-005	Rental	Fulfillment	Completion	01 Jan 26	Active
RR-006	F&B	Sale	Immediate	01 Jan 26	Active

Status

Active

◦ Draft

Pending Approval

Expired

● Inactive

Row actions

View

Edit

Clone

Deactivate

Assign Products

View Transactions

View Audit

Important

Rules need:

- Unique rule code
- Description
- Effective From
- Effective To
- Version
- Status
- Owner
- Approval status

The matrix specifically requires rule creation, modification, activation/deactivation, unique codes, validity periods, version history, cloning and audit.

Screen 3 — Create / Edit Revenue Recognition Rule

This is where Finance defines **when revenue becomes earned**.

Section A — Basic Information

Rule Code

RR-001

Rule Name

Standard Admission — Recognition on Scan

Description

Recognize admission revenue when ticket is first successfully validated.

Status

Active

Effective From

01 Jan 2026

Effective To

Optional

Section B — Recognition Trigger

Dropdown:

Recognition Event

Possible options should include:

- Sale Payment
- Ticket Usage
- First Usage
- Redemption
- Performance Booking
- Ticket Expiry
- Validity Date
- Advance Payment
- Deposit Collection
- Deposit Fulfillment
- Wallet Load
- Wallet Usage
- Wallet Cash-out
- Wallet Expiry
- External Wallet Transaction
- Credit Settlement
- Credit Write-off

- Scheduled Date

These come directly from the matrix's revenue-recognition scenarios.

Example

Trigger:

First Successful Ticket Scan

Section C — Recognition Method

Options:

Immediate

100% on Trigger

Fixed Percentage

Time Based

Usage Based

Daily Allocation

Monthly Allocation

Validity-Based Allocation

Custom Rule

Example:

Recognition Method:

100% on First Usage

Section D — Accounting

Deferred Account:

210100 — Deferred Ticket Revenue

Revenue Account:

410100 — Admission Revenue

Tax Treatment:

VAT already recognized at sale

Suspense Account:

999999 — Revenue Suspense

Section E — Controls

Prevent Duplicate Recognition: **Yes**

Allow Manual Override: **Finance Manager Only**

Requires Approval for Change: **Yes**

Retroactive Rule Changes: **No**

Buttons

Save Draft

Validate Rule

Submit for Approval

Save & Activate

Screen 4 — Revenue Allocation / Split Builder

This screen handles products where one sale contains value belonging to multiple attractions/products/entities.

This is particularly important for TICVAI because your matrix explicitly requires configurable revenue splits for multi-attraction and combination tickets.

Example**UAE Explorer Pass**

Selling Price:

AED 500

Contains:

Aquarium

Observation Deck

Museum

Transport

Allocation Method

Dropdown:

Fixed Percentage

Other options:

Fixed Amount

Percentage

Usage Based

First Usage

Redemption Based

Relative Product Value

Custom

Allocation

Component	Allocation	Amount
Aquarium	35%	AED 175
Observation Deck	30%	AED 150
Museum	20%	AED

Component	Allocation	Amount
		100
Transport	15%	AED 75

Bottom:

Total Allocation

100%

AED 500

Valid

If it becomes:

Aquarium 35%

Observation Deck 30%

Museum 20%

Transport 10%

Total = **95%**

TICVAI shows:

Allocation must equal 100%.

Save should be blocked.

More advanced rules

Finance could also configure:

Recognize Aquarium allocation only when Aquarium is visited.

Meaning the AED 175 remains deferred until that attraction is used.

This is a very important difference between **allocation** and **recognition**.

Screen 5 — Product-to-Revenue Rule Mapping

Board 1 mapped **transactions to GL accounts**.

This page maps **products to revenue-recognition rules**.

Left hierarchy

Admissions

- ↳ Aquarium
 - ↳ Adult Ticket
 - ↳ Child Ticket
 - ↳ VIP Ticket
- ↳ Observation Deck
- ↳ Museum

Membership

- ↳ Annual Pass
- ↳ Monthly Pass

Wallet**Gift Cards****Rental****Table**

Product	Rule	Deferred Account	Revenue Account	Status
Aquarium Adult	RR-001	210100	410100	Active
Aquarium Child	RR-001	210100	410100	Active
Annual Membership	RR-002	210200	420100	Active
Wallet	RR-003	220100	Depends on Use	Active

Assignment levels

Allow Finance to assign a rule at:

Product Group

↓

Product

↓

Product Variant / Matrix Cell

This is directly aligned with the matrix requirements for Matrix Sheets, Matrix Cells, product attributes, bulk assignment and mass updates.

Bulk action

Finance selects 250 admission products.

Assign RR-001

instead of editing 250 products individually.

Screen 6 — Deferred Revenue Management

This should give Finance direct visibility into money TICVAI has collected but has **not yet earned**.

Header**Deferred Revenue****KPI cards**

Total Deferred

AED 1,185,300

Ticket Deferred

AED 420,000

Membership Deferred

AED 560,000

Wallet Liability

AED 150,000

Other

AED 55,300

Main table

Product	Collected	Recognized	Deferred	Next Recognition
Annual Membership	1,200,000	640,000	560,000	31 Aug
Admission Tickets	980,000	560,000	420,000	On Usage
Wallet	350,000	200,000	150,000	On Redemption

Drill-down

Click:

Admission Tickets — AED 420,000

Then show individual deferred transactions.

PNR	Product	Sale Date	Deferred	Trigger	Expiry
PNR-1001	Adult Ticket	10 Aug	250	Usage	30 Sep
PNR-1002	VIP Ticket	10 Aug	500	Usage	30 Sep

Aging

Add:

0–30 Days

31–60 Days

61–90 Days

90+ Days

This addresses the matrix's deferred-revenue aging requirement.

Screen 7 — Revenue Recognition Scheduler

Not every recognition event happens immediately.

For memberships, annual passes, multi-day products and some accounting policies, Finance needs scheduled recognition.

Header

Recognition Schedules

Example

Annual Membership

Customer paid:

AED 1,200

Membership:

12 Months

Recognition:

AED 100/month

Timeline

Jan — AED 100 ✓

Feb — AED 100 ✓

Mar — AED 100 ✓

...

Dec — AED 100

Schedule types

Daily

Weekly

Monthly

Yearly

Custom Frequency

Validity Duration

Create Schedule

Schedule Name

Rule

Frequency

Start Date

End Date

Run Time

Timezone

Company

Product Group

Example

Membership Monthly Recognition

Frequency:

Monthly

Execution:

Last Day of Month — 23:30

Actions

Pause

Resume

Run Now

Edit

View History

The matrix explicitly requires daily, weekly, monthly, yearly and configurable recognition scheduling, including automated background jobs and manual execution.

Screen 8 — Revenue Recognition Job Monitor

Screen 7 defines **when jobs should run**.

Screen 8 shows **what actually happened when they ran**.

Header

Revenue Recognition Jobs

KPIs

Jobs Today

28

Successful

25

Running

1

Failed

2

Revenue Recognized

AED 640,000

Table

Job	Rule	Transactions	Amount	Started	Status
JOB-8201	Membership Monthly	4,281	420,000	00:01	Success
JOB-8202	Ticket Expiry	1,245	185,000	01:00	Success
JOB-8203	Wallet Expiry	340	35,000	02:00	Failed

Open job

JOB-8203 — Wallet Expiry

Transactions processed: 340

Successful: 329

Failed: 11

Recognized: AED 32,500

Failed: AED 2,500

Failure reasons

7 Missing GL Mapping

3 Invalid Wallet Status

1 Closed Fiscal Period

Actions

View Failed Transactions

Retry Failed

Export

View Journal Entries

This is critical operationally because Finance must know whether automated recognition actually succeeded.

Screen 9 — Revenue Recognition Exceptions

This is where Finance resolves recognition problems.

Header**Revenue Exceptions****KPI cards**

Open Exceptions

31

Financial Impact

AED 185,420

Critical

7

Older Than 24h

5

Exception categories

Missing Revenue Rule

Missing GL Mapping

Allocation ≠ 100%

Duplicate Recognition

Invalid Product Mapping

Closed Fiscal Period

Recognition Job Failed

ERP Rejection

Suspense Revenue

Table

Transaction	Product	Amount	Problem	Age	Status
TIC-90821	VIP Ticket	1,500	Missing Rule	2h	Open
TIC-90832	Combo Pass	4,500	Allocation 95%	5h	Critical
TIC-90845	Membership	12,000	Closed Period	1d	Open

Open exception

Show:

Transaction

TIC-90832

Product

UAE Explorer Pass

Amount

AED 4,500

Problem

Allocation totals 95%.

System recommendation

Missing allocation:

5% / AED 225

Actions

Correct Rule

Correct Mapping

Move to Suspense

Retry Recognition

Assign Owner

Escalate

Important principle

Again, TICVAI should **never silently fix financial errors**.

It should detect them, explain them and route them to authorized Finance users.

Screen 10 — Revenue Recognition Audit, Reports & Traceability

The final page brings the entire revenue-recognition lifecycle together.

Search

Search:

PNR

Transaction

Product

Recognition Rule

Journal

Customer

Example

Search:

PNR-874251

Revenue timeline

11 Aug — 14:21

Ticket sold

AED 500

↓

AED 500 → Deferred Revenue

Rule:

RR-001

↓

11 Aug — 16:32

Ticket scanned

↓

Recognition Trigger:

First Usage

↓

AED 500 recognized

↓

Journal:

JE-100422

↓

DR Deferred Revenue AED 500

CR Admission Revenue AED 500

↓

ERP:

SAP-8745103

Posted

Reports section

Provide reports for:

- Revenue Recognition
- Deferred Revenue
- Revenue Allocation
- Product Revenue
- Site Revenue
- Attraction Revenue
- GL Posting
- Suspense Revenue
- Revenue Reconciliation
- Rule Audit

These are directly supported by the reporting requirements in the matrix.

Export

Excel

CSV

PDF

ERP Export

Scheduled Export

Important special cases Board 3 must support

1. Normal Ticket

Customer pays AED 200.

At Sale

Cash/Card → AED 200

Deferred Revenue → AED 200

Customer scans ticket.

At Usage

Deferred Revenue → AED 200

Admission Revenue → AED 200

2. Expired Unused Ticket

Customer pays AED 200 but never comes.

Sale:

AED 200 → Deferred Revenue

Ticket expires.

Expiry rule triggers:

AED 200 Deferred

→

AED 200 Earned Revenue

This requirement appears explicitly in your matrix.

3. Annual Membership

Customer pays:

AED 1,200

Do not necessarily recognize AED 1,200 immediately.

Deferred:

AED 1,200

Then:

AED 100 Jan

AED 100 Feb

AED 100 Mar

...

depending on the configured accounting policy.

4. Digital Wallet

Customer loads:

AED 1,000

This is **not revenue yet**.

It becomes:

Cash/Card +AED 1,000

Wallet Liability +AED 1,000

Then customer spends AED 150 at F&B:

Wallet Liability -AED 150

F&B Revenue +AED 150

Remaining wallet liability:

AED 850

Your matrix specifically requires wallet revenue to be realized on redemption rather than initial top-up.

5. Combo Ticket

Customer pays:

AED 500

Aquarium → AED 175

Observation Deck → AED 150

Museum → AED 100

Transport → AED 75

Each component may potentially have **its own recognition event**.

For example:

Aquarium visited:

AED 175 recognized.

Museum not yet visited:

AED 100 remains deferred.

That's why the Revenue Engine needs to work at the **allocation/component level**, not just at Order level.

One architectural recommendation for Board 3

Separate these three concepts

1. ACCOUNT MAPPING

Where does the money go?

Example:

Deferred Ticket Revenue → GL 210100

2. REVENUE ALLOCATION

Who does the money belong to?

Example:

AED 500 Combo Pass

→ Aquarium AED 175

→ Observation Deck AED 150

→ Museum AED 100

→ Transport AED 75

3. REVENUE RECOGNITION

When does that money become earned revenue?

Example:

Aquarium AED 175

→ Recognized when Aquarium ticket is scanned.

These should **not be combined into one hard-coded rule**.

The architecture should therefore be:

Product / Transaction

↓

Financial Event

↓

Allocation Engine

↓

Deferred Revenue

↓

Recognition Rules Engine

↓

Recognition Event

↓

Double-Entry Journal

↓

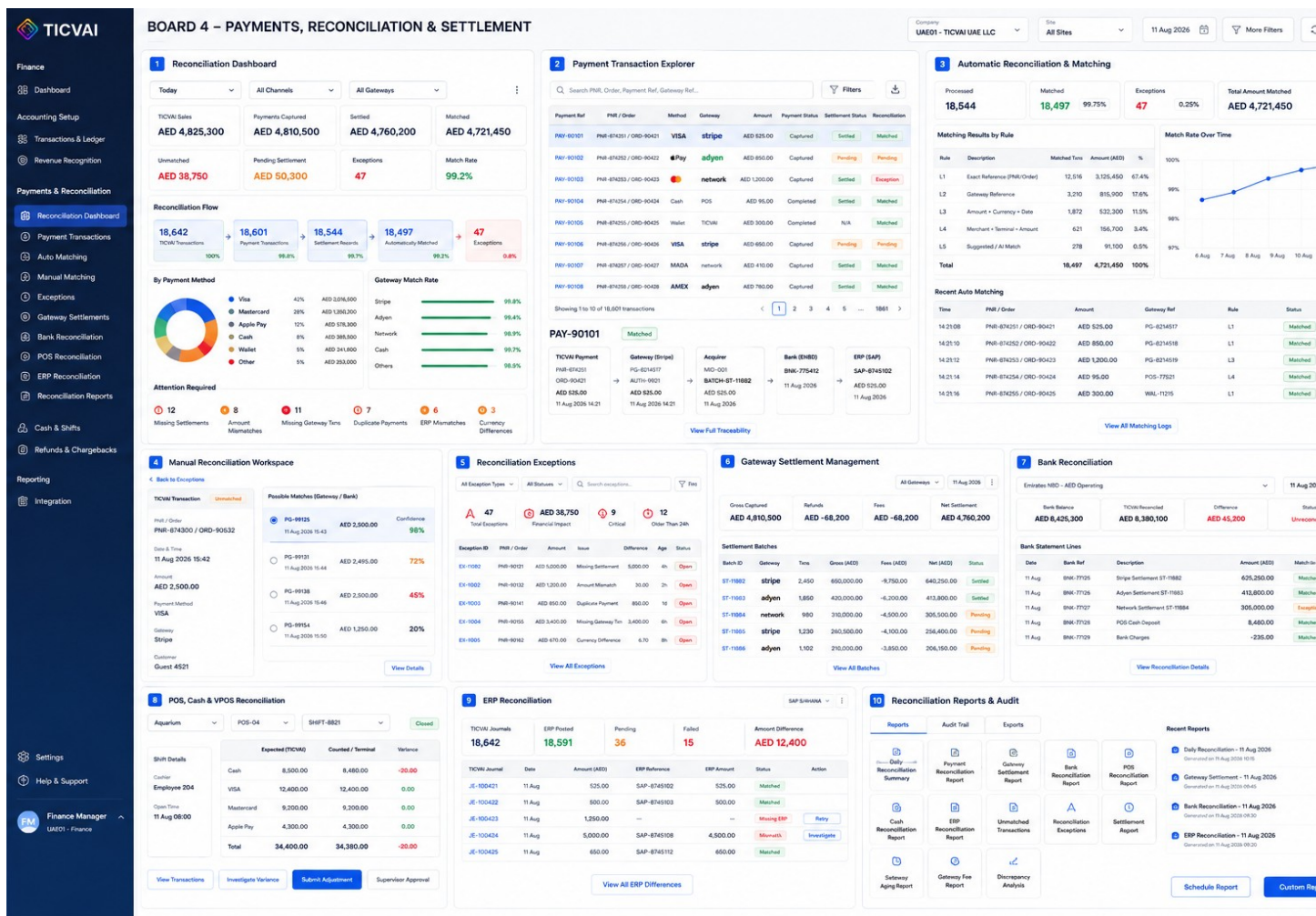
General Ledger



ERP

This separation will give TICVAI much more flexibility for attractions, memberships, wallets, packages, multi-attraction tickets and future products. It also matches the depth of the Revenue Recognition requirements already present in your matrix

BOARD 4 — Payments, Reconciliation & Settlement



Purpose

Board 2 answered:

What accounting transaction was created?

Board 3 answered:

When and how is revenue recognized?

Board 4 answers:

Did we actually receive the correct money, and does every system agree?

The core flow should be:

TICVAI Sale → Payment Gateway/POS → Acquirer → Settlement → Bank → Reconciliation → ERP

Screen 1 — Reconciliation Dashboard

This is the Finance team's daily reconciliation control center.

Global filters

Company: UAE01

Site: All Sites

Channel: All

Gateway: All

Acquirer: All

Payment Method: All

Currency: AED

Settlement Date: 11 Aug 2026

Main KPI cards

TICVAI Sales

AED 4,825,300

Payments Captured

AED 4,810,500

Settled

AED 4,760,200

Matched

AED 4,721,450

Unmatched

AED 38,750

Pending Settlement

AED 50,300

Exceptions

47

Match Rate

99.2%

Reconciliation status

Show a visual funnel:

18,642 TICVAI Transactions

↓

18,601 Payment Transactions

↓

18,544 Settlement Records

↓

18,497 Automatically Matched

↓

47 Exceptions

This immediately tells Finance where the problem exists.

Payment method split

Visa

Mastercard

Amex

Apple Pay

Cash

Wallet

Voucher

Other

Gateway status

Example:

Stripe — 99.8% Matched

Adyen — 99.4%

Network — 98.9%

Cash — 99.7%

Attention Required

12 Missing Settlements

8 Amount Mismatches

11 Missing Gateway Transactions

7 Duplicate Payments

6 ERP Mismatches

3 Currency/FX Differences

Every number should be clickable.

Screen 2 — Payment Transaction Explorer

This is the master list of payments rather than accounting journals.

Header

Payment Transactions

Search:

PNR / Order / Payment Reference / Gateway Reference / Customer / Card Last 4

Filters

- Date
- Company
- Site
- Channel
- Payment Method
- Gateway
- Acquirer
- Currency
- Payment Status
- Settlement Status

- Reconciliation Status

Table

Payment Ref	PNR	Method	Gateway	Amount	Payment	Settlement	Reconciliation
PAY-90101	PNR-874251	Visa	Stripe	AED 525	Captured	Settled	Matched
PAY-90102	PNR-874252	Apple Pay	Adyen	AED 850	Captured	Pending	Pending
PAY-90103	PNR-874253	Mastercard	Network	AED 1,200	Captured	Settled	Exception

Open payment

Show:

TICVAI Payment

Order: TIC-90421

PNR: PNR-874251

Amount: AED 525

Gateway

Gateway Ref: PG-8214517

Authorization: AUTH-99121

Acquirer

Merchant ID: MID-001

Settlement Batch: ST-11882

Bank

Bank Reference: BNK-775412

ERP

ERP Reference: SAP-8745102

The Finance user can therefore see the **entire payment chain on one screen**.

Screen 3 — Automatic Reconciliation & Matching

This screen controls TICVAI's automatic reconciliation engine.

Header

Automatic Matching

Today's results

Transactions Processed: **18,544**

Automatically Matched: **18,497**

Match Rate: **99.75%**

Exceptions: **47**

Matching rules

The engine should attempt matching using:

Level 1 — Exact Reference

PNR / Order Reference

↓

Level 2 — Gateway Reference

↓

Level 3 — Amount + Currency + Date

↓

Level 4 — Merchant + Terminal + Amount

↓

Level 5 — Intelligent Suggested Match**Example**

TICVAI:

PNR-874251

AED 525

11 Aug 14:21

Gateway:

PG-8214517

PNR-874251

AED 525

11 Aug 14:21

Result:

100% Match

Another example

TICVAI:

AED 1,000

Gateway:

AED 970

Result:

Amount Mismatch — AED 30

Possible explanation:

Gateway fee AED 30.

But TICVAI should **not automatically assume that** unless the configured reconciliation rule confirms it.

Matching rule configuration

Allow Finance to configure:

- Exact Reference Required
- Amount Tolerance
- Date Tolerance
- Currency Matching

- Gateway Fee Treatment
- Partial Settlement
- Batch Settlement
- Split Settlement

Screen 4 — Manual Reconciliation Workspace

Not every transaction will match automatically.

This page is where Finance resolves them.

Layout recommendation

Use **two sides**.

LEFT — TICVAI

Transaction:

PNR-874300

Date: 11 Aug 2026

Amount: AED 2,500

Payment: Visa

Gateway: Stripe

Merchant: MID-001

RIGHT — Gateway / Bank

Possible Match #1

PG-99125

AED 2,500

11 Aug 2026

Confidence:

98%

Possible Match #2

PG-99131

AED 2,495

11 Aug 2026

Confidence:

72%

Actions

Match

Partial Match

Split Match

Mark Exception

Investigate

Recommendation

I would add a **Match Confidence Score**.

Example:

98% Suggested Match

Reasons:

- ✓ Same amount
- ✓ Same date
- ✓ Same merchant
- ✓ Same card last 4
- △ PNR missing

Finance can then approve the match.

This is a good place to use AI/advanced matching without allowing AI to silently change financial records.

Screen 5 — Reconciliation Exceptions

This is the investigation queue.

KPI cards

Open Exceptions

47

Financial Impact

AED 38,750

Critical

9

Older Than 24 Hours

12

Exception types

Missing Gateway Transaction

Missing TICVAI Transaction

Amount Mismatch

Duplicate Payment

Duplicate Settlement

Partial Settlement

Currency Difference

Gateway Fee Difference

Refund Mismatch

Chargeback

ERP Mismatch

Unknown Payment

Table

Exception	PNR	Amount	Issue	Difference	Age	Owner
EX-1001	PNR-90121	5,000	Missing Settlement	5,000	4h	Finance
EX-1002	PNR-90132	1,200	Amount Mismatch	30	2h	Sara
EX-1003	PNR-90141	850	Duplicate	850	1d	Omar

Exception detail

Show:

Expected

vs.

Actual

Example:

TICVAI

AED 1,200

Gateway

AED 1,170

Difference

AED 30

Potential reason:

Processing Fee

Actions

Resolve

Match

Create Adjustment

Assign Owner

Escalate

Add Note

Attach Evidence

Screen 6 — Gateway Settlement Management

A payment being captured does not necessarily mean the money has reached the merchant's bank account.

This page tracks gateway/acquirer settlement.

Header**Gateway Settlements****KPIs**

Captured Today

AED 4.81M

Settled

AED 4.76M

Pending

AED 50.3K

Gateway Fees

AED 68.2K

Settlement batches

Batch	Gateway	Transactions	Gross	Fees	Net	Status
ST-11882	Stripe	2,450	650,000	9,750	640,250	Settled
ST-11883	Adyen	1,850	420,000	6,200	413,800	Settled
ST-11884	Network	980	310,000	4,500	305,500	Pending

Open batch

Gross Transactions

AED 650,000

Refunds

-AED 12,000

Chargebacks

-AED 3,000

Fees

-AED 9,750

Expected Settlement

AED 625,250

Actual Settlement:

AED 625,250

Matched

Drill-down

Finance should be able to see every payment included in the settlement batch.

Screen 7 — Bank Reconciliation

Now we compare settlement records against actual bank activity.

Header**Bank Reconciliation****Bank accounts**

Example:

Emirates NBD — AED Operating Account

Bank Balance:

AED 8,425,300

TICVAI Reconciled:

AED 8,380,100

Difference:

AED 45,200

Bank statement lines

Date	Bank Ref	Description	Amount	Match
11 Aug	BNK-7712 5	Stripe Settlement	625,250	Matched
11 Aug	BNK-7712 6	Adyen Settlement	413,800	Matched
11 Aug	BNK-7712 7	Network Payment	305,000	Exception

TICVAI expected

Network Settlement:

AED 305,500

Bank received:

AED 305,000

Difference:

AED 500

Finance can investigate.

Actions

Match

Find Settlement

Create Adjustment

Mark Bank Fee

Investigate

Screen 8 — POS, Cash & VPOS Reconciliation

Your matrix specifically asks for easy automated reconciliation of card payments through VPOS/POS against ticketing payments.

This screen reconciles by:

Site → POS → Till → Shift → Payment Type

Example

Aquarium — POS 04

Shift:

SHIFT-8821

Cashier:

Employee 204

TICVAI expected

Cash: AED 8,500

Visa: AED 12,400

Mastercard: AED 9,200

Apple Pay: AED 4,300

Total: AED 34,400

POS / Terminal

Cash Counted: AED 8,480

Visa Terminal: AED 12,400

Mastercard: AED 9,200

Apple Pay: AED 4,300

Difference

Cash:

-AED 20

Card:

AED 0

Total:

-AED 20

Actions

View Shift

View Transactions

Investigate Variance

Submit Adjustment

Supervisor Approval

This connects Board 4 directly to the Cash/Shift board later.

Screen 9 — ERP Reconciliation

This screen compares TICVAI's financial records against what was successfully posted into the client's ERP.

Header**ERP Reconciliation****KPI cards**

TICVAI Journals:

18,642

ERP Posted:

18,591

Pending:

36

Failed:

15

Financial Difference:

AED 12,400

Table

TICVAI Journal	Amount	ERP Ref	TICVAI	ERP	Status
JE-100421	525	SAP-8745102	525	525	Matched
JE-100422	500	SAP-8745103	500	500	Matched
JE-100423	1,250	—	1,250	—	Missing ERP
JE-100424	5,000	SAP-8745108	5,000	4,500	Mismatch

Open difference

TICVAI:

AED 5,000

SAP:

AED 4,500

Difference:

AED 500

Possible causes

GL Mapping

Tax Mapping

Currency Conversion

Cost Center

Partial ERP Posting

Posting Rejection

Actions

View Journal

View ERP Payload

Retry Posting

Create Adjustment

Escalate

This closes the loop:

Operational transaction → TICVAI Finance → ERP → Reconciliation

Screen 10 — Reconciliation Reports & Audit

The final screen gives Finance and auditors complete reconciliation reporting.

Main reports

Daily Reconciliation Summary

Payment Reconciliation Report

Gateway Settlement Report

Bank Reconciliation Report

POS Reconciliation Report

Cash Reconciliation Report

ERP Reconciliation Report

Unmatched Transactions

Reconciliation Exceptions

Gateway Fee Report

Settlement Aging

Example Daily Reconciliation

11 August 2026

TICVAI Sales:

AED 4,825,300

Payments Captured:

AED 4,810,500

Gateway Settled:

AED 4,760,200

Bank Received:

AED 4,759,700

ERP Posted:

AED 4,747,300

Outstanding

Pending Gateway:

AED 50,300

Bank Difference:

AED 500

ERP Difference:

AED 12,400

Open Exceptions:

47

Audit trail

Every manual match should show:

Who matched it

When

Original records

Reason

Supporting attachment

Approval

Adjustment journal

ERP impact

Nothing should disappear after reconciliation.

One very important feature I would add

Reconciliation Waterfall

We would advise making a visual **money journey** available for any transaction.

For example:

PNR-874251

Customer Paid

AED 525

↓

TICVAI

AED 525 ✓

↓

Payment Gateway

AED 525 ✓

↓

Acquirer

AED 525 ✓

↓

Gateway Fee

-AED 7.88

↓

Expected Bank Settlement

AED 517.12

↓

Bank Received

AED 517.12 ✓

↓

TICVAI Journal

AED 525 ✓

↓

ERP

AED 525 ✓

Result**Fully Reconciled**

This is much easier for a Finance Manager to understand than looking through five separate tables.

For a problematic transaction:

Customer Paid

AED 1,000

↓

TICVAI

AED 1,000 ✓

↓

Gateway

AED 1,000 ✓

↓

Expected Bank

AED 985

↓

Actual Bank

AED 980

↓

Difference AED 5

Finance immediately knows **where the discrepancy occurred**.

Another architectural point

Just like Revenue Recognition, reconciliation should **not be coded separately inside every payment channel**.

There should be a central:

Reconciliation Engine

Inputs:

TICVAI Transactions

•

Payment Gateway

•

Acquirer

•

POS/VPOS

•

Cash/Till

•

Bank

•

ERP

↓

Normalization Layer

All sources converted into a common reconciliation record:

Reference → Source → Amount → Currency → Date → Merchant → Payment Method → Settlement → Status

↓

Matching Engine

Exact Matching

Rule-Based Matching

Tolerance Matching

Suggested/AI Matching

↓

Exceptions

↓

Finance Resolution

↓

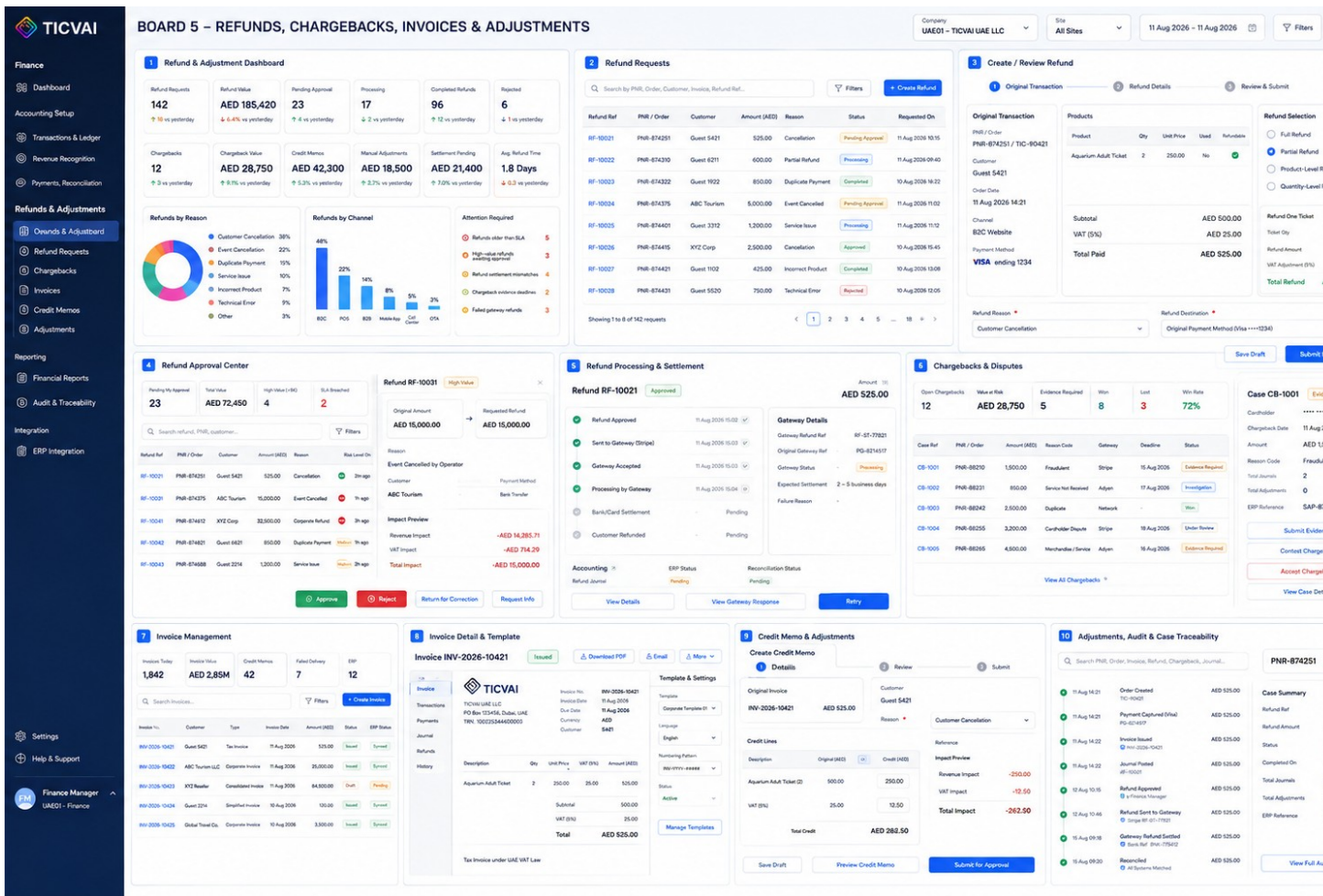
Adjustment / Approval

↓

Fully Reconciled

This architecture is important because different TICVAI clients may use completely different combinations—one customer may use Stripe + SAP, another Network + Oracle, and another Adyen + Microsoft Dynamics. The core reconciliation process should remain the same while the connectors change. That is consistent with the reconciliation, ERP and payment requirements already defined in your matrix.

BOARD 5 — Refunds, Chargebacks, Invoices & Adjustments



Purpose

The basic flow is:

Original Transaction → Refund / Dispute / Adjustment → Approval → Financial Treatment → Payment/Gateway Action → Journal → Reconciliation → ERP → Audit

I recommend the following 10 screens.

Screen 1 — Refund & Adjustment Dashboard

This is the Finance Manager's overview of all post-sale financial activity.

Global filters

Company: UAE01

Site: All Sites

Attraction: All

Channel: All

Payment Method: All

Period: August 2026

KPI cards**Refund Requests**

142

Refund Value

AED 185,420

Pending Approval

23

Processing

17

Completed Refunds

96

Rejected

6

Chargebacks

12

Chargeback Value

AED 28,750

Additional financial KPIs**Credit Memos**

AED 42,300

Manual Adjustments

AED 18,500

Refund Settlement Pending

AED 21,400

Average Refund Time

1.8 Days

Charts**Refunds by Reason**

Customer Cancellation

Event Cancellation

Duplicate Payment

Service Issue

Incorrect Product

Technical Error

Other

Refunds by channel

B2C

POS

B2B

Mobile App

Call Center

OTA

Attention Required

5 Refunds older than SLA

3 High-value refunds awaiting approval

4 Refund settlement mismatches

2 Chargeback evidence deadlines

3 Failed gateway refunds

Every alert should open the related cases.

Screen 2 — Refund Request Management

This is the master list of refund requests.

The matrix specifically requires tracking the complete lifecycle including **request, approval, processing, settlement, completion, rejection and cancellation**, together with reason, approver, payment method, status, timestamps and financial impact.

Header

Refund Requests

Button:

+ Create Refund

Search

Search by:

PNR

Order

Ticket

Customer

Payment Reference

Invoice

Refund Reference

Filters

Refund Status

Refund Reason

Site

Channel

Payment Method

Requested By

Amount

Date

Table

Refund	PNR	Customer	Original	Refund	Reason	Status
RF-10021	PNR-874251	Guest 5421	525	525	Cancellation	Pending Approval
RF-10022	PNR-874310	Guest 6211	1,200	600	Partial Refund	Processing
RF-10023	PNR-874322	Guest 1922	850	850	Duplicate Payment	Completed

Status workflow

Draft

→ Submitted

→ Pending Approval

→ Approved

→ Processing

→ Settlement Pending

→ Completed

or:

Rejected

Cancelled

Failed

This status model should be standardized across TICVAI.

Screen 3 — Create / Review Refund

This is where the actual refund is prepared.

Step 1 — Original Transaction

Search:

PNR-874251

System retrieves:

Customer: Guest 5421

Order: TIC-90421

Order Date: 11 Aug 2026

Channel: B2C

Payment: Visa

Original Amount: AED 525

Invoice: INV-2026-10421

Products

Product	Qty	Price	Used?	Refundable?
---------	-----	-------	-------	-------------

Aquarium Adult	2	AED 250	No	Yes
----------------	---	---------	----	-----

VAT:

AED 25

Total:

AED 525

Step 2 — Select refund

Options:

Full Refund**Partial Refund****Product-Level Refund****Quantity-Level Refund**

Example:

Refund one ticket:

AED 250

VAT adjustment:

AED 12.50

Total Refund: AED 262.50

Step 3 — Refund reason

Mandatory dropdown:

Customer Cancellation

Event Cancellation

Duplicate Charge

Incorrect Product

Service Failure

Operational Closure

Technical Error

Goodwill

Other

If "Other":

mandatory comment.

Step 4 — Refund destination

Default:

Original Payment Method

Visa ending 1234

Potential options depending on policy:

Original Card

Wallet Credit

Voucher

Bank Transfer

Cash

Step 5 — Financial preview

Before submitting, TICVAI should show:

Original

Revenue AED 250

VAT AED 12.50

↓

Refund Impact

Revenue -AED 250

VAT Liability -AED 12.50

↓

Customer Refund

AED 262.50

Buttons

Save Draft

Submit for Approval

Screen 4 — Refund Approval Center

Refund approval should use the same **central TICVAI Approval Engine** we discussed for Finance journals and other platform approvals.

Header**Refund Approvals****KPIs**

Pending My Approval

23

Total Value

AED 72,450

High Value

4

SLA Breached

2

Table

Refund	Amount	Reason	Requested By	Risk	Age
RF-10021	525	Cancellation	Agent 102	Low	20m
RF-10032	15,000	Event Cancelled	Supervisor	High	1h
RF-10041	32,500	Corporate Refund	Manager	High	3h

Approval detail

Show:

Original Transaction

AED 15,000

**Requested Refund**

AED 15,000



Products affected



Revenue impact



VAT impact



Payment destination



Reason & supporting evidence

Approval rules

Example:

Below AED 500

→ Automatic / Supervisor

AED 500–5,000

→ Finance Manager

AED 5,000–25,000

→ Finance Manager + Department Head

Above AED 25,000

→ CFO

These values should be **configurable**, not hard-coded.

Actions

Approve

Reject

Return for Correction

Request Information

Add Comment

Screen 5 — Refund Processing & Settlement

Approval does **not** mean the customer has received the money.

This page tracks what happens after approval.

Header

Refund Processing

Example

RF-10021

Refund:

AED 525

Payment:

Visa

Gateway:

Stripe

Original Gateway Ref:

PG-8214517

Timeline

Refund Approved

✓ 11 Aug 15:02

↓

Sent to Gateway

✓ 11 Aug 15:03

↓

Gateway Accepted

✓ 11 Aug 15:03

↓

Processing

↓

Bank/Card Settlement

Pending

↓

Customer Refunded

Pending

Gateway information

Gateway Refund Reference:

REF-ST-77821

Refund Amount:

AED 525

Gateway Status:

Processing

Expected Settlement:

2–5 business days

Accounting

Journal:

RF-JE-10028

ERP:

Pending

Reconciliation:

Pending

Exception example

If gateway returns:

Refund Failed — Original payment method unavailable

Finance gets:

Retry

Change Refund Method

Escalate

View Gateway Response

Important

This screen should connect directly to **Board 4 reconciliation**, because TICVAI needs to confirm that the refund actually appeared in gateway/bank settlement.

Screen 6 — Chargeback & Dispute Management

This is a completely different process from a normal refund.

The matrix explicitly requires **chargeback notification, dispute management, evidence collection, investigation workflows, resolution tracking and financial adjustments**, including chargeback status, amount, payment method, gateway and outcome.

Header**Chargebacks & Disputes****KPIs**

Open Chargebacks

12

Value at Risk

AED 28,750

Evidence Required

5

Won

8

Lost

3

Win Rate

72%

Table

Case	PNR	Amount	Reason	Gateway	Deadline	Status
CB-1001	PNR-88210	1,500	Fraud	Stripe	15 Aug	Evidence Required
CB-1002	PNR-88231	850	Service Not Received	Adyen	17 Aug	Investigation
CB-1003	PNR-88242	2,500	Duplicate	Network	—	Won

Open case

Chargeback CB-1001

Customer

Order

Tickets

Payment

Gateway

Card last 4

Transaction Date

Chargeback Date

Amount

Reason Code

Evidence section

Finance/Operations can attach:

Invoice

Ticket

Booking Confirmation

Access/Scan Record

Customer Communication

Terms & Conditions

Proof of Service

Important TICVAI advantage

Because TICVAI also controls ticket validation, the system can automatically show:

Ticket was successfully scanned at Aquarium Gate 3 on 7 Aug 2026 at 16:42.

That can become valuable dispute evidence.

Workflow

Received

→ Investigation

→ Evidence Required

→ Submitted

→ Under Review

→ Won / Lost

Actions

Submit Evidence

Accept Chargeback

Contest

Assign Owner

Screen 7 — Invoice Management

The matrix requires TICVAI to support **tax invoices, simplified invoices, corporate invoices and consolidated invoices**, PDF generation, email delivery, ERP integration, numbering, templates and multilingual formats.

Header**Invoices**

Button:

+ Create Invoice

KPI cards

Invoices Today

1,842

Invoice Value

AED 2.85M

Credit Memos

42

Failed Delivery

7

ERP Pending

12

Filters

Invoice Type

Company

Customer

Site

Channel

Status

Tax Status

ERP Status

Table

Invoice	Customer	Type	Amount	VAT	Status	ERP
INV-10421	Guest 5421	Tax Invoice	525	25	Issued	Synced
INV-10422	ABC Tourism	Corporate	25,000	1,190	Issued	Synced
INV-10423	XYZ Reseller	Consolidated	84,500	4,024	Draft	—

Invoice types

Tax Invoice

Simplified Tax Invoice

Corporate Invoice

Consolidated Invoice

Actions

View

Download PDF

Email

Resend

Create Credit Memo

View Transaction

View ERP

Screen 8 — Invoice Detail & Template

Opening an invoice should show the actual document plus the transaction behind it.

Invoice header

TICVAI Client Company

Tax Registration Number

Address

Invoice Number

Invoice Date

Customer

Customer TRN if applicable

Currency

Lines

Description	Qty	Unit Price	VAT	Total
Aquarium Adult Ticket	2	250	25	525

Subtotal

AED 500

VAT

AED 25

Total

AED 525

Language

English | العربية | Bilingual

Related records

Order

Payment

PNR

Journal

Refund

Credit Memo

ERP Reference

Template configuration

Authorized admin should be able to manage:

Logo

Company Details

VAT Fields

Header/Footer

Legal Text

Language

Numbering Pattern

Example:

INV-UAE-AQ-2026-00010421

Important

Invoice numbering should be controlled and auditable. Issued invoice numbers should not simply disappear because somebody deleted a transaction.

Screen 9 — Credit Memo & Financial Adjustment

This handles formal reductions/corrections linked to existing invoices and transactions.

The matrix explicitly requires full and partial credit memos linked to invoices, refunds, cancellations and adjustments with audit trails and financial traceability.

Header

Create Credit Memo

Original Invoice

INV-10421

AED 525

Credit type

Full Credit

Partial Credit

Product Credit

Tax Adjustment

Price Correction

Service Adjustment

Example

Original Invoice:

AED 525

Credit:

AED 262.50

Remaining Invoice Value:

AED 262.50

Credit memo lines

Product	Original	Credit
Adult Ticket	500	250
VAT	25	12.50

Financial preview

Revenue adjustment:

-AED 250

VAT adjustment:

-AED 12.50

Customer balance:

-AED 262.50

Relationship

The UI should visibly show:

INV-10421

↓

CM-10082

↓

RF-10021

↓

Journal JE-10821

↓

ERP SAP-92105

This gives full financial traceability.

Actions

Save Draft

Submit for Approval

Issue Credit Memo

Screen 10 — Adjustments, Audit & Case Traceability

The final screen should give Finance one place to investigate the entire lifecycle of any refund, invoice, chargeback or adjustment.

Global search

Search:

PNR

Order

Invoice

Refund

Chargeback

Credit Memo

Journal

Payment

Customer

Example search

PNR-874251

Timeline

11 Aug — 14:21

Order Created

AED 525

↓

Payment Captured

Visa

↓

Invoice INV-10421 Issued

↓

Journal JE-100421 Posted

↓

12 Aug — 10:15

Refund RF-10021 Requested

↓

Finance Approval

↓

Refund Sent to Stripe

↓

Credit Memo CM-10082 Created

↓

Refund Journal RF-JE-10028

↓

ERP Updated

↓

Gateway Refund Settled

↓

Bank Reconciled

↓

Final status

Refund Fully Completed & Reconciled

Audit details

Every event should show:

- Date/time
- User/system
- Action
- Original value
- New value
- Reason
- Approval
- Attachment
- Financial impact
- Journal
- ERP reference

One important workflow

This simple example will explain Board 5 very well.

Customer purchases:

2 Tickets = AED 500 + AED 25 VAT

Total:

AED 525

Later, customer requests refund for **one ticket**.

TICVAI should automatically understand:

Original transaction:

2 tickets

↓

Select:

1 ticket

↓

Eligible refund:

AED 250

↓

VAT reversal:

AED 12.50

↓

Customer refund = AED 262.50

↓

Approval if required

↓

Payment gateway refund

↓

Credit Memo

↓

Accounting reversal/adjustment

↓

ERP

↓

Reconciliation

This should be **one connected workflow**.

Finance should not manually calculate the VAT, manually create a credit memo, separately request the gateway refund and then manually create an accounting journal.

Another important recommendation — Refund Policy Engine

I would add a centralized **Refund Policy Engine**.

This goes slightly beyond the explicit matrix wording, but it is a strong product recommendation because TICVAI will have many different products and clients.

Example:

Standard Ticket

More than 48 hours before visit:

100% refundable

24–48 hours:

50% refundable

Below 24 hours:

Non-refundable

VIP Experience

Cancellation before 72 hours:

100%

Below 72 hours:

Non-refundable

Event Cancelled by Operator

Automatic 100% refund

Membership

Refund:

Finance approval required

Then when an agent selects:

PNR-874251

TICVAI automatically says:

Refund Eligibility

Eligible

Maximum refundable:

AED 262.50

Policy:

Standard Ticket — More than 48h

Approval:

Not required

Or:

Not normally refundable

Reason:

Ticket already redeemed.

Manager Override Required

This would make refunds much safer and easier to operate.

Architectural recommendation for Board 5

Development should **not** build Refunds, Chargebacks, Credit Memos and Adjustments as disconnected processes.

They should use a common:

Financial Case Engine

A financial case contains:

Case Type

Refund / Chargeback / Credit Memo / Adjustment

↓

Original Financial Transaction

↓

Reason

↓

Amount

↓

Policy / Eligibility

↓

Approval

↓

Financial Impact

↓

Payment Action

↓

Journal

↓

ERP

↓

Reconciliation

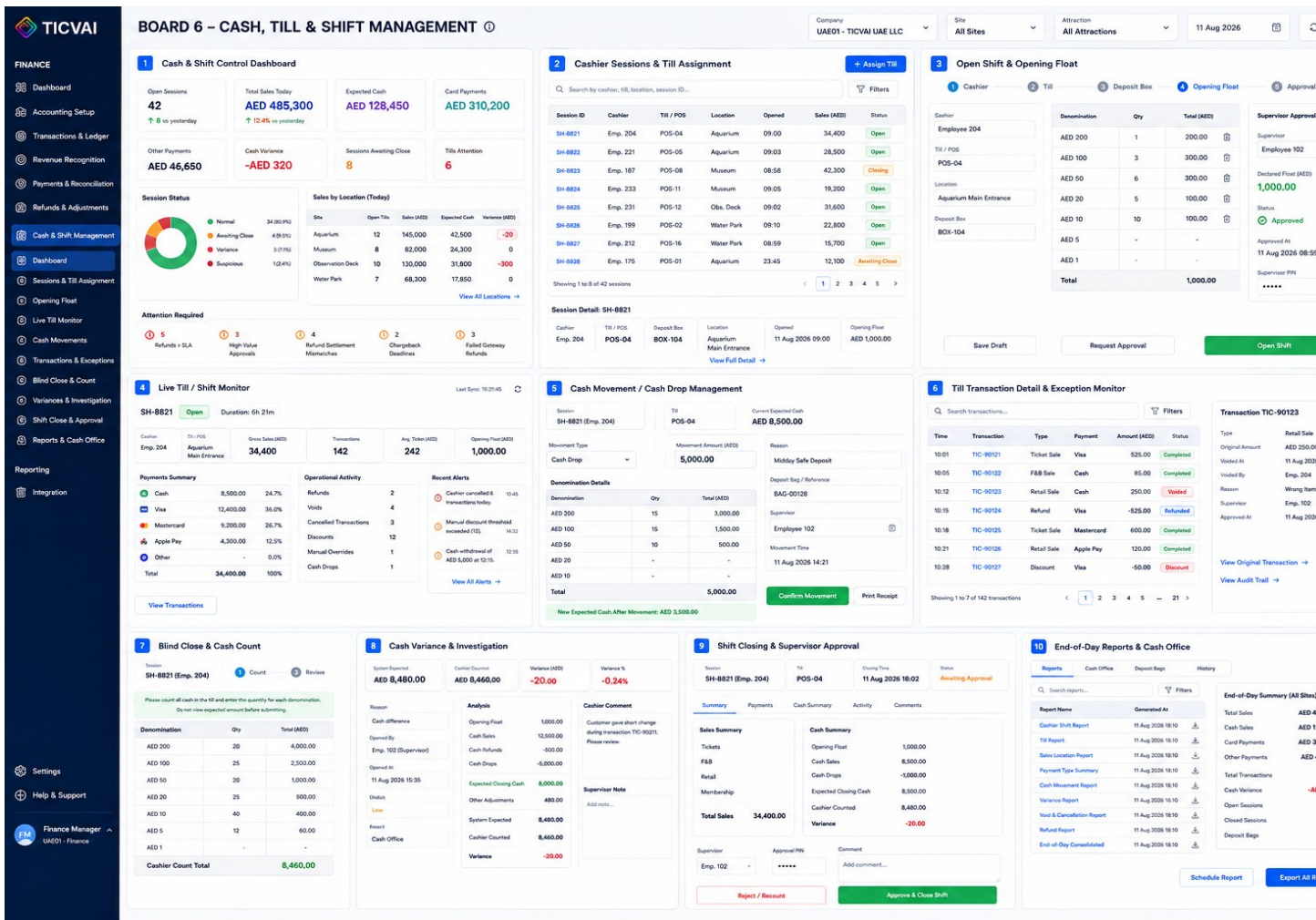
↓

Audit

That gives TICVAI one standardized way to manage financial corrections and disputes.

And importantly, **Board 5 should connect directly to Boards 2, 3 and 4**: a refund may reverse/reclassify a journal from Board 2, reverse recognized/deferred revenue from Board 3, and then require payment/bank reconciliation through Board 4. That end-to-end connection is essential for the accounting-grade finance layer described throughout your matrix.

BOARD 6 — Cash, Till & Shift Management



Purpose

The complete flow should be:

Cashier → Till Assignment → Opening Float → Shift → Sales → Cash Movements → Monitoring → Closing Count → Variance → Approval → Deposit → Reconciliation → Finance

I recommend these **10 backend pages**.

Screen 1 — Cash & Shift Control Dashboard

This is the Finance/Cash Office/Supervisor overview of all tills and cashier sessions.

Global filters

Company: UAE01

Site: All Sites

Attraction: All

Sales Location: All

Date: 11 Aug 2026

Additional filters:

Open | Closed | Variance | Attention Required

KPI cards**Open Cashier Sessions**

42

Total Sales Today

AED 485,300

Expected Cash

AED 128,450

Card Payments

AED 310,200

Other Payments

AED 46,650

Cash Variance

-AED 320

Tills Requiring Attention

6

Sessions Awaiting Close

8

Session status overview

34 Normal

4 Awaiting Close

3 Variance

1 Suspicious Activity

Live locations

Site	Open Tills	Sales	Expected Cash	Variance
Aquarium	12	145,000	42,500	-20
Museum	8	82,000	24,300	0
Observation Deck	10	130,000	31,800	-300

Clicking Aquarium opens its tills.

Attention Required

Examples:

POS-04 cash variance AED 20

POS-12 session open for 14 hours

POS-08 awaiting supervisor close

Cashier 204 cancelled 6 transactions

POS-17 unusual cash withdrawal

This should be another **exception-driven dashboard**.

Screen 2 — Cashier Sessions & Till Assignment

This screen manages who is operating which POS/till.

Header

Cashier Sessions

Button:

+ Assign Till

Table

Session	Cashier	Till	Location	Opened	Sales	Status
SH-8821	Employee 204	POS-04	Aquarium	09:00	34,400	Open
SH-8822	Employee 221	POS-05	Aquarium	09:03	28,500	Open
SH-8823	Employee 187	POS-08	Museum	08:58	42,300	Closing

Session detail

Cashier

Employee 204

Till

POS-04

Deposit Box

BOX-104

Location

Aquarium Main Entrance

Opening Time

09:00

Opening Float

AED 1,000

Important control

Your matrix says the system should only allow **one session per user**.

Therefore:

If Employee 204 tries to open POS-07 while POS-04 is still open:

Active Session Exists

Employee 204 currently has an active session on POS-04.

Options for authorized supervisor:

View Session

Force Close

Transfer Session

But everything must be audited.

Screen 3 — Open Shift & Opening Float

This controls how a cashier starts their shift.

Step 1 — Cashier

Employee 204

Step 2 — Till

POS-04

Step 3 — Deposit Box

BOX-104

Step 4 — Opening Float

The system should support **two methods**.

Method A — Total Amount

Cashier enters:

AED 1,000

or:

Method B — Count by Denomination

Denominatio n	Qt y	Tota l
AED 200	1	200
AED 100	3	300
AED 50	6	300
AED 20	5	100
AED 10	10	100

Total Float = AED 1,000

The matrix specifically requires denomination-level counting and saving each denomination count.

Supervisor control

Depending on configuration:

Cashier Count:

AED 1,000

Supervisor Declared Float:

AED 1,000

Match

Then:

Supervisor PIN / Approval

Actions

Save Draft

Request Approval

Open Shift

Configuration

The client should decide whether:

Cashier can open independently

or

Supervisor approval required

Your matrix explicitly requires configurable supervisor approval for opening/closing cashier sessions.

Screen 4 — Live Till / Shift Monitor

Once the shift is open, supervisors need live visibility.

Example

SH-8821 — POS-04

Cashier:

Employee 204

Status:

Open

Duration:

6h 21m

Sales

Gross Sales:

AED 34,400

Transactions:

142

Average Transaction:

AED 242

Payments

Cash:

AED 8,500

Visa:

AED 12,400

Mastercard:

AED 9,200

Apple Pay:

AED 4,300

Operational activity

Refunds: 2

Voids: 4

Cancelled Transactions: 3

Discounts: 12

Manual Overrides: 1

Cash Drops: 1

Timeline

09:00 Shift Opened

09:01 Float AED 1,000

10:42 Transaction voided

12:15 Cash withdrawal AED 5,000

13:10 Refund AED 250

14:32 Supervisor override

Alerts

If unusual activity happens:

Cashier cancelled 5 transactions within 20 minutes.

or:

Manual discount activity exceeds configured threshold.

This doesn't automatically mean fraud—it means the supervisor should review it.

Screen 5 — Cash Movement / Cash Drop Management

During the day, the supervisor may remove money from the cashier's till.

Your matrix specifically requires the supervisor to be able to withdraw cash from the cashier float during the day and trace it.

Header

Cash Movement

Movement types

Cash Drop

Cash Withdrawal

Float Increase

Float Decrease

Cash Transfer

Safe Deposit

Petty Cash

depending on client configuration.

Example

Till:

POS-04

Cashier:

Employee 204

System Expected Cash:

AED 8,500

Supervisor wants to remove:

AED 5,000

Denominations

AED 200 × 15 = 3,000

AED 100 × 15 = 1,500

AED 50 × 10 = 500

Total:

AED 5,000

After movement

Expected till cash:

AED 8,500

Less cash drop:

-AED 5,000

New expected cash:

AED 3,500

Control

Supervisor:

Employee 102

Reason:

Midday Safe Deposit

Timestamp:

14:21

Deposit Bag:

BAG-00128

Actions

Confirm

Print Receipt

Send to Cash Office

The system should create an immutable cash-movement record.

Screen 6 — Till Transaction Detail & Exception Monitor

Your matrix says a detailed log of transactions for each till must always be accessible, searchable and printable from back office.

Header**Till Transactions — POS-04****Search/filter**

Transaction

PNR

Payment

Product

Amount

Cashier

Transaction Type

Time

Status

Table

Time	Transaction	Type	Payment	Amount	Status
10:01	TIC-90121	Ticket	Visa	525	Completed
10:05	TIC-90122	F&B	Cash	85	Completed
10:12	TIC-90123	Retail	Cash	250	Voided
10:15	TIC-90124	Refund	Visa	-525	Approved

Highlight unusual records

Voided

Cancelled

Refunded

Manual Override

Open transaction

Show:

Original Order

Products

Payment

Cashier

Customer

Void/Cancel Reason

Supervisor Approval

Journal

Audit

Notifications

The matrix specifically says controlling teams should receive notifications when sessions open/close and when transactions are cancelled by cashiers.

So notification rules should be configurable.

Screen 7 — Blind Close & Cash Count

This is one of the most important Cash Control screens.

What is Blind Close?

The cashier should **not see what TICVAI expects them to have** before counting.

Otherwise, if the system says:

Expected Cash:

AED 8,480

the cashier could simply enter:

AED 8,480.

Instead, the cashier sees:

Close Shift

Count your cash

Denomination count

Denomination	Qty	Total
AED 200	20	4,000
AED 100	25	2,500
AED 50	20	1,000
AED 20	25	500
AED 10	40	400
AED 5	12	60

Cashier Count

AED 8,460

Then:

Submit Count

Only after submission does TICVAI reveal:

System Expected:

AED 8,480

Cashier Count:

AED 8,460

Difference:

-AED 20 SHORT

This directly supports the matrix requirement for blind close and recording over/short values.

Screen 8 — Cash Variance & Investigation

Any over/short should go into a controlled investigation workflow.

Header

Cash Variances

KPIs

Open Variances

12

Total Short

-AED 820

Total Over

+AED 230

Net Variance

-AED 590

Table

Shift	Cashier	Expected	Counted	Variance	Status
SH-8821	Employee 204	8,480	8,460	-20	Review
SH-8825	Employee 231	12,200	12,250	+50	Review
SH-8830	Employee 192	5,400	5,000	-400	Critical

Open variance

System Expected:

AED 8,480

Cashier Count:

AED 8,460

Difference:

-AED 20

Supporting analysis

Opening Float: AED 1,000

Cash Sales: +AED 12,480

Refunds: -AED 500

Cash Drops: -AED 4,500

Expected: AED 8,480

Cashier comment

Customer change correction during transaction TIC-90211.

Supervisor

Can investigate:

Transaction history

Voids

Refunds

Cash movements

Opening count

Closing count

Actions

Accept Variance

Request Recount

Investigate

Escalate

Create Adjustment

Threshold rules

AED 0–20

→ Supervisor

AED 20–100

→ Cash Office Manager

Above AED 100

→ Finance Manager

Again, thresholds should be configurable.

Screen 9 — Shift Closing & Supervisor Approval

Once cash is counted and variances handled, the shift can be formally closed.

Header**Shift Close — SH-8821****Shift summary**

Cashier:

Employee 204

POS:

POS-04

Opening:

09:00

Closing:

18:02

Sales summary

Tickets:

AED 18,500

F&B:

AED 6,400

Retail:

AED 5,200

Membership:

AED 4,300

Gross Sales: AED 34,400

Payment summary

Cash:

AED 8,500

Visa:

AED 12,400

Mastercard:

AED 9,200

Apple Pay:

AED 4,300

Cash control

Opening Float:

AED 1,000

Cash Sales:

AED 8,500

Cash Drop:

-AED 1,000

Expected Closing Cash:

AED 8,500

Counted:

AED 8,480

Variance:

-AED 20

Operational summary

Transactions: 142

Voids: 4

Refunds: 2

Cancelled: 3

Discounts: 12

Supervisor action

Approve Close

Reject / Recount

Add Comment

Escalate Variance

Once approved:

SHIFT CLOSED

The till becomes available for another authorized session.

Automatic close

Your matrix also requires automatic session closing after a configurable period with supervisor notification.

Example:

Session SH-8821 has exceeded maximum 14-hour duration.

Then system can:

Auto Close / Lock

depending on policy.

Screen 10 — End-of-Shift / End-of-Day Reports & Cash Office

The final page consolidates cashier activity into Finance reporting.

Header**Cash & Shift Reports****Report options****Cashier Shift Report****Till Report****Sales Location Report****Payment Type Report****Cash Movement Report****Variance Report****Void & Cancellation Report****Refund Report****Deposit Box Report****End-of-Day Consolidated Report****Example — Cashier Shift Report****Employee 204**

SH-8821

11 Aug 2026

Product summary

Admission Tickets:

74 units — AED 18,500

F&B:

88 items — AED 6,400

Retail:

21 items — AED 5,200

Membership:

4 — AED 4,300

Payment summary

Cash AED 8,500

Visa AED 12,400

Mastercard AED 9,200

Apple Pay AED 4,300

Activity

Transactions: 142

Voids: 4

Refunds: 2

Cancelled: 3

Cash

Opening Float: AED 1,000

Cash Drop: AED 1,000

Closing Expected: AED 8,500

Counted: AED 8,480

Variance: -AED 20

Comments

Cashier comments

Supervisor comments

Output

PDF

Excel

CSV

Print

Email

The matrix specifically requires shift reports, temporary closing reports with payment-type and denomination totals, comments, and automatic consolidated daily payment reports sent to desired teams.

Deposit-box transaction outputs should support **Arabic and English**, as required by the matrix.

Sample workflow**09:00 — Opening**

Cashier receives:

AED 1,000 Float

↓

Supervisor approves

↓

During Day

Cash sales:

+AED 12,500

↓

Supervisor removes:

-AED 5,000 cash drop

↓

Cash refund:

-AED 500

↓

Expected Closing Cash

AED 8,000

↓

Blind Count

Cashier counts:

AED 7,980

↓

Variance

AED 20 Short

↓

Cashier comment

↓

Supervisor review

↓

Variance approved

↓

Shift closed

↓

Cash transferred to Cash Office

↓

Bank deposit

↓

Board 4 Reconciliation

This creates a continuous financial chain rather than disconnected cashier functions.

Important recommendation — Cash Custody Chain

I would add this because the matrix covers the cashier/deposit box well, but we should extend the design to answer an important question:

What happens to physical cash after it leaves the cashier?

The system should be able to trace:

Cashier

↓

Till

↓

Cash Drop

↓

Deposit Bag / Deposit Box

↓

Supervisor / Cash Office

↓

Safe

↓

Bank Deposit

↓

Bank Confirmation

↓

Reconciliation

For example:

BAG-00128

Cashier: Employee 204

Till: POS-04

Amount: AED 5,000

Handed By: Employee 204

Received By: Supervisor 102

Cash Office Received: 14:42

Safe: SAFE-01

Bank Deposit:

DEP-88210

Bank Received:

AED 5,000

Status:

Fully Reconciled

This gives large attractions a proper **chain of custody for physical cash**.

Architectural recommendation for Board 6

Development team not to build separate logic for every POS.

There should be a common:

Cash & Shift Engine

Core object:

Cashier Session

↓

Till / POS

↓

Deposit Box

↓

Opening Float

↓

Transactions

↓

Payment Totals

↓

Cash Movements

↓

Blind Count

↓

Variance

↓

Approval

↓

Closing

↓

Cash Custody

↓

Reconciliation

And the same engine should work whether the cashier is selling:

Tickets

F&B

Retail

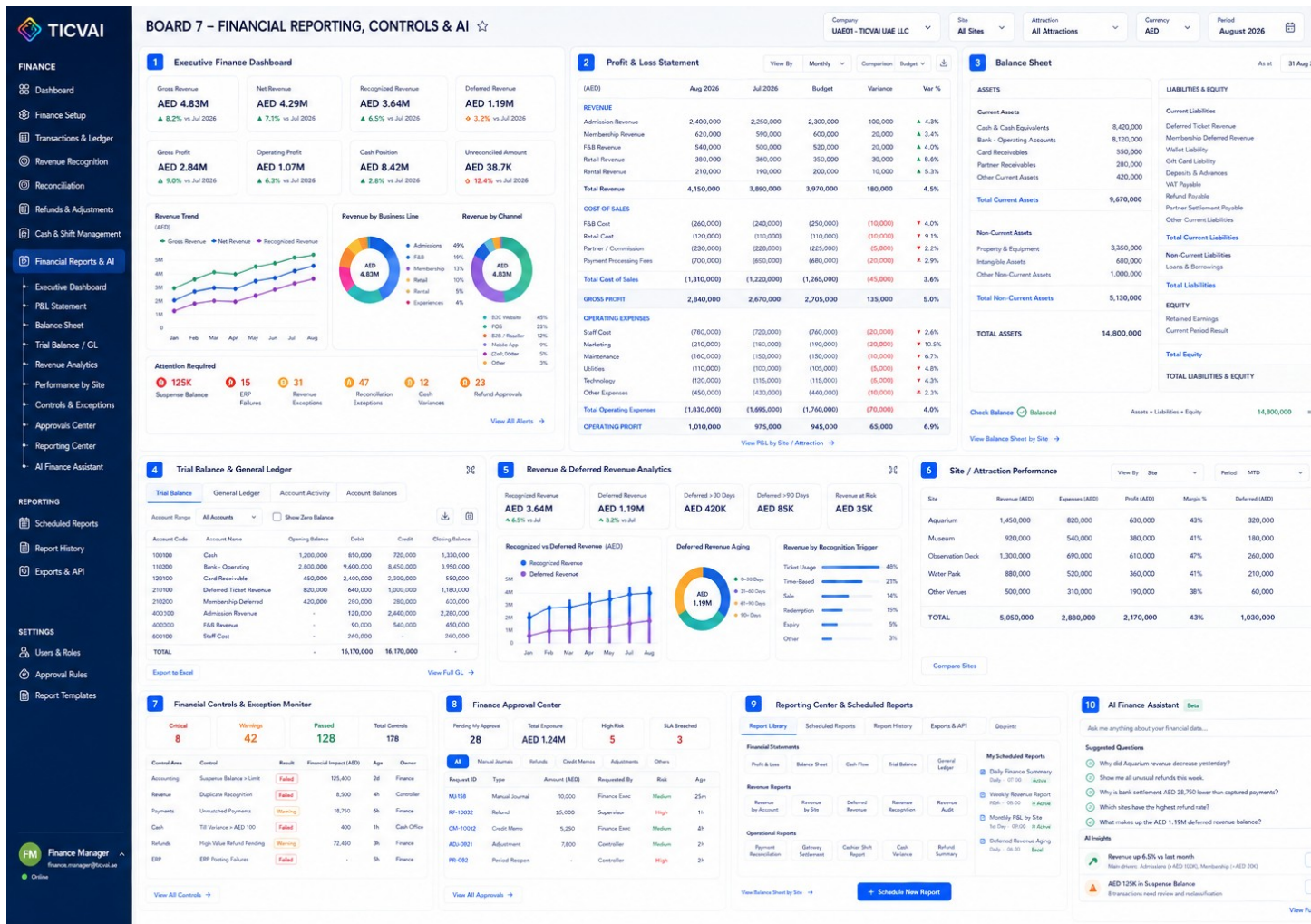
Membership

Rental

or several of them from the same unified POS.

That last point is particularly important for TICVAI: because the POS is intended to support multiple business lines, **the cash/till engine should belong to the platform—not to Ticketing, F&B, or Retail individually.**

BOARD 7 — Financial Reporting, Controls & AI



Purpose

Board 7 answers:

“What is the financial position, what changed, where are the risks, and what action should Finance take?”

The flow is:

Ledger + Revenue + Payments + Cash + Refunds + Reconciliation

↓

Financial Reporting

↓

Controls & Exceptions

↓

Approvals

↓

Executive Analysis

↓

AI Insights

I recommend these **10 pages**.

Screen 1 — Executive Finance Dashboard

This should be the top-level CFO/Finance Manager view.

Global filters

Company: UAE01

Site: All Sites

Attraction: All

Currency: AED

Period: August 2026

Comparison: Previous Month / Previous Year / Budget

KPI cards

Gross Revenue

AED 4.83M

Recognized Revenue

AED 3.64M

Net Revenue

AED 4.29M

Deferred Revenue

AED 1.19M

Gross Profit

AED 2.84M

Refunds

AED 185K

Cash Position

AED 8.42M

Unreconciled

AED 38.7K

Profitability KPIs

Gross Margin

66.2%

Operating Margin

24.8%

Revenue Growth

+8.2%

Refund Rate

3.8%

Reconciliation Match Rate

99.2%

Financial trend

Revenue vs Expense vs Profit

Jan → Aug

Revenue by business

Admissions

Membership

F&B

Retail

Rental

Experiences

Revenue by channel

B2C

POS

B2B

OTA

Mobile

Call Center

Attention Required

AED 125K Suspense Balance

15 ERP Posting Failures

31 Revenue Exceptions

47 Reconciliation Exceptions

12 Open Cash Variances

23 Refund Approvals

This should be highly drillable.

Clicking **AED 125K Suspense** opens the exact transactions.

Screen 2 — Profit & Loss Statement

This is the main operating financial statement.

Header

Profit & Loss

Filters:

Company

Site

Attraction

Department

Period

Currency

Comparison

Example structure

Revenue

Admission Revenue

AED 2,400,000

Membership Revenue

AED 620,000

F&B Revenue

AED 540,000

Retail Revenue

AED 380,000

Rental Revenue

AED 210,000

Total Revenue

AED 4,150,000

Cost of Sales

F&B Cost

Retail Cost

Partner Costs

Commission

Payment Processing Fees

Total Cost of Sales

AED 1,310,000

Gross Profit

AED 2,840,000

Operating Expenses

Staff

Marketing

Maintenance

Utilities

Technology

Other

Operating Profit**AED 1,030,000****Columns**

Actual

Previous Period

Budget

Variance

Variance %

Important

Finance should be able to drill down:

P&L → Revenue Line → GL Account → Journal → Original Transaction

For example:

Admission Revenue AED 2.4M

click

↓

Admission Revenue GL

↓

12,420 journal entries

↓

Ticket transactions

This creates proper traceability.

Screen 3 — Balance Sheet

The matrix explicitly requires Balance Sheet reporting.

Header**Balance Sheet****Assets**

Cash & Cash Equivalents

Bank

Card Receivables

Partner Receivables

Other Current Assets

Liabilities

Deferred Ticket Revenue

Membership Deferred Revenue

Wallet Liability

Gift Card Liability

Deposits

VAT Payable

Refund Payable

Partner Settlement Payable

Equity

Retained Earnings

Current Period Result

Example

Assets

AED 14.8M

Liabilities

AED 6.2M

Equity

AED 8.6M

Validation:

Assets = Liabilities + Equity

Balanced

Important drill-down

Click:

Wallet Liability — AED 420,000

↓

Show:

Number of wallets

Wallet balances

Unredeemed funds

Expiry

Related GL

Transactions

This is much more valuable than just a static financial statement.

Screen 4 — Trial Balance & General Ledger Reports

The matrix explicitly requires Trial Balance and General Ledger reports.

Trial Balance

Account	Openin g	Debi t	Credi t	Closing
Cash	1.2M	850K	720K	1.33M
Card Receivable	450K	2.4M	2.3M	550K

Account	Openin g	Debi t	Credi t	Closing
Deferred Revenue	820K	640K	1.0M	1.18M
Admission Revenue	—	120K	2.4M	2.28M

Validation

Total Debit

AED 12,540,000

Total Credit

AED 12,540,000

Balanced

Filters

Account Range

Account Type

Company

Site

Cost Center

Profit Center

Period

Currency

Tabs**Trial Balance****General Ledger****Account Activity****Account Balances**

This consolidates multiple reporting requirements from the matrix.

Screen 5 — Revenue & Deferred Revenue Analytics

Board 3 handled the operational process.

This page provides senior-level analytics.

KPI cards

Recognized Revenue

AED 3.64M

Deferred Revenue

AED 1.19M

Deferred >30 Days

AED 420K

Deferred >90 Days

AED 85K

Revenue at Risk

AED 35K

Charts

Recognized vs Deferred

By month

Deferred Revenue Aging

0–30

31–60

61–90

90+

Revenue by product

Admission

Membership

Wallet

Gift Card

Packages

Rental

Revenue by recognition trigger

Sale

Usage

Redemption

Expiry

Time

Drill-down example

AED 85K >90 Days

↓

Show products

↓

Transactions

↓

Reason not recognized

This helps Finance identify potentially stale or incorrectly configured revenue.

Screen 6 — Site, Attraction & Business Performance

Your matrix includes site-level reporting, balance reporting, revenue tracking, P&L reporting and cross-site financial consolidation.

This page compares businesses.

Table

Site	Revenue	Expenses	Profit	Margin	Deferred	Refund Rate
Aquarium	1.45M	820K	630K	43%	320K	2.8%
Museum	920K	540K	380K	41%	180K	3.2%
Observation Deck	1.30M	690K	610K	47%	260K	2.4%

Charts

Revenue by site

Profit margin by site

Revenue per visitor

Average spend

Refund rate

Payment fee %

Deferred revenue %

Compare mode

Select:

Aquarium vs Museum

Then TICVAI shows:

Revenue

Attendance

Average Ticket Value

Revenue per Guest

Refund %

Gateway Fees

Operating Margin

This would be very useful for management.

Screen 7 — Financial Controls & Exception Monitor

This is one of the pages I would strongly recommend beyond pure reporting.

Finance needs a consolidated control screen.

Header**Financial Controls****Control categories****Accounting**

Unbalanced Journals

Manual Journals

Backdated Journals

Reopened Periods

Suspense Entries

Revenue

Missing Rules

Duplicate Recognition

Allocation Errors

Old Deferred Revenue

Payments

Unmatched Payments

Settlement Differences

Duplicate Payments

Cash

Till Variances

Excessive Voids

High Cash Drops

Refunds

High-Value Refunds

Refunds After Usage

Repeated Refunds

Example**Critical Controls — 8****Warnings — 42****Passed Controls — 128****Table**

Control	Result	Financial Impact	Owner	Age
Suspense Balance > Limit	Failed	125,400	Finance	2d
Duplicate Recognition	Failed	8,500	Controller	4h
Cash Variance > AED 100	Failed	400	Cash Office	1h

Control	Result	Financial Impact	Owner	Age
Closed Period Posting	Warning	15,000	Finance	2h

Actions

Investigate

Assign

Resolve

Accept Exception

Escalate

Recommendation

Allow clients to configure control thresholds.

Example:

Alert if Suspense Balance > AED 50,000

or

Alert if refunds exceed 5% of daily revenue.

Screen 8 — Finance Approval Center

We have approval pages inside journals/refunds/etc., but Board 7 needs a **central finance approval inbox**.

Header**My Finance Approvals****KPI cards**

Pending My Approval

28

Total Financial Exposure

AED 1.24M

High Risk

5

SLA Breached

3

Approval types

Manual Journal

Journal Reversal

Refund

Credit Memo

Revenue Rule Change

Fiscal Period Reopen

Cash Variance

Financial Adjustment

High-Value Discount

Commission Settlement

Table

Request	Type	Amount	Requested By	Risk	Age
MJ-158	Manual Journal	10,000	Finance Exec	Medium	25m
RF-10032	Refund	15,000	Supervisor	High	1h
FP-08	Period Reopen	—	Controller	High	2h

Open approval

Show:

Current



Requested Change



Financial Impact



Policy / Rule



Supporting Documents



Audit history

Actions

Approve

Reject

Return for Correction

Delegate

Add Comment

Important

Again, this should be the **same common TICVAI approval engine**, not a Finance-only approval engine built separately.

Screen 9 — Reporting Center & Scheduled Reports

The matrix explicitly includes CSV/Excel exports, scheduled exports, financial-report exports and API-based data extraction.

Header**Financial Reporting Center****Report library****Financial Statements**

Profit & Loss

Balance Sheet

Cash Flow

Trial Balance

General Ledger

Revenue

Revenue by Account

Revenue by Site

Deferred Revenue

Revenue Recognition

Revenue Allocation

Payments

Payment Reconciliation

Gateway Settlement

Bank Reconciliation

Operations

Cashier Shift

Cash Variance

Refunds

Chargebacks

Report scheduler

Example:

Daily Finance Summary

Frequency:

Daily

Time:

07:00

Recipients:

Finance Team

Format:

PDF + Excel

Scope:

All UAE Sites

Scheduling options

Daily

Weekly

Monthly

Quarterly

Custom

Delivery

Email

Download Center

SFTP

API

ERP

Report history

Show:

Generated Date

Status

Size

Recipients

Download

Recommendation

Each report should have a **Save View** function.

For example:

“Dubai Aquarium Monthly P&L”

with predefined filters.

Screen 10 — AI Finance Assistant

This should be one of TICVAI's major differentiators.

Not just:

“Show me revenue.”

But:

“Explain why the revenue changed.”

Main screen

Ask TICVAI Finance AI

Example questions:

Why did Aquarium revenue fall yesterday?

Show me all unusual refunds this week.

Why is the bank settlement AED 38,750 lower than TICVAI payments?

Which sites have the highest refund rate?

What makes up the AED 1.18M deferred revenue balance?

Show all manual journal entries above AED 50,000 this month.

Compare August profitability between Aquarium and Museum.

Example AI response

Question:

Why is yesterday's bank settlement AED 38,750 lower than captured payments?

TICVAI Finance AI

Captured payments:

AED 4,810,500

Bank settled:

AED 4,771,750

Difference:

AED 38,750

Breakdown:

Pending Gateway Settlement

AED 21,500

Refunds

AED 8,250

Chargebacks

AED 4,000

Gateway Fees

AED 3,500

Unmatched Transactions

AED 1,500

Result

Difference fully explained:

AED 38,750

Buttons:

View 14 Transactions

Open Reconciliation

Export Analysis

This would be genuinely useful.

Another AI example — Profitability

Finance Manager asks:

Why did Museum profitability decrease this month?

TICVAI responds:

Revenue:

+3.2%

but

Operating Expenses:

+14.5%

Main drivers:

Staff Overtime

+AED 42,000

Promotional Discounts

+AED 28,000

Payment Processing Fees

+AED 12,500

Refunds

+AED 8,200

Net impact:

Operating Margin decreased from 44% to 38%.

Then:

View Cost Breakdown

Important AI control

I would be very strict here.

AI should be allowed to:

Search

Analyze

Explain

Compare

Detect anomalies

Recommend

Forecast

Create draft reports

Suggest reconciliation matches

AI should NOT autonomously:

Post journals

Approve refunds

Change GL mappings

Change revenue rules

Reopen accounting periods

Issue credit memos

Transfer money

Approve cash variances

Those actions must go through normal permissions and approvals.

AI can say:

"I recommend reclassifying these 12 transactions."

Then:

[Create Draft Reclassification]

But Finance still reviews and approves it.

One important page (To be considered)

To have a dedicated page for the below.

Cash Flow Statement

Your matrix specifically calls for Cash Flow Statement support together with P&L, Balance Sheet, Trial Balance and GL reporting.

For Board 7 now, Cash Flow can sit inside **Screen 9 — Reporting Center**, but later it could become its own page if needed.

Architectural recommendation for Board 7

Development team to build a common:

Finance Analytics Layer

Not separate reporting queries for every dashboard.

All finance modules should feed a governed finance analytics model:

Transactions

-

Ledger

-

Revenue Recognition

-

Payments

-

Reconciliation

-

Refunds

-

Cash

↓

Finance Analytics Layer

Dimensions:

Company

Site
 Attraction
 Product
 Channel
 Customer
 Account
 Cost Center
 Profit Center
 Time
 Currency
 Payment Method
 ↓

Outputs

P&L
 Balance Sheet
 Trial Balance
 Revenue Analytics
 Site Performance
 Dashboards
 Reports
 AI

This is especially important for the AI assistant. If dev team lets the AI directly query dozens of operational tables independently, the answers may become inconsistent. The AI should query a governed **financial semantic/analytics layer** using the same definitions used by Finance reporting.

That way:

“Revenue”

means the same thing in the P&L, executive dashboard, report export, and AI answer.

That consistency is critical for an enterprise financial system.

And with Board 7, the main Finance backend structure is complete:

Board 1 — Finance Setup & COA

Board 2 — Transactions & Ledger

Board 3 — Revenue Recognition

Board 4 — Reconciliation & Settlement

Board 5 — Refunds, Chargebacks & Invoices

Board 6 — Cash & Shift Management

Board 7 — Reporting, Controls & AI

Together, these form one connected Finance layer rather than seven disconnected modules.

